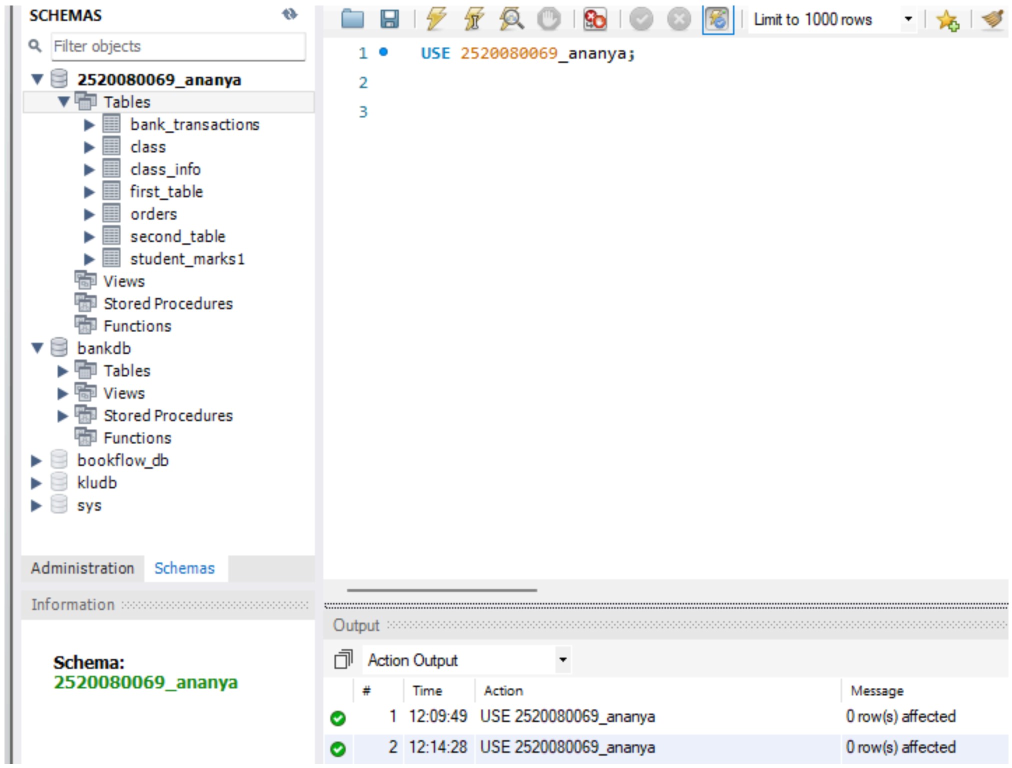
SKILL WEEK-4 : ACID PROPERTIES AND ISOLATION

===

1. CREATE DATABASE

===

CREATE DATABASE Bank_ACID_DB;
USE Bank_ACID_DB;

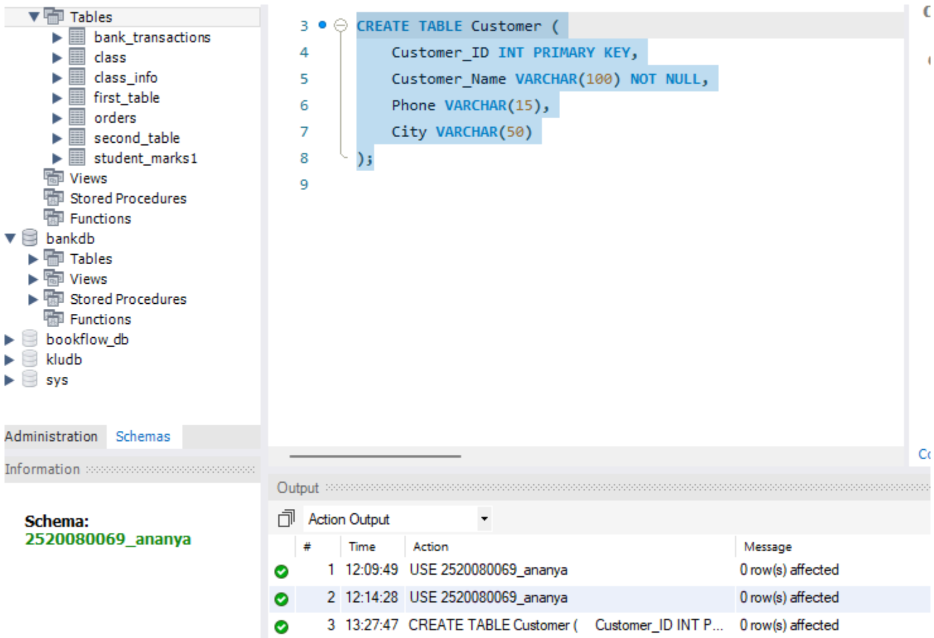


===

2. CREATE CUSTOMER TABLE

===

```
CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Customer_Name VARCHAR(100) NOT NULL,
    Phone VARCHAR(15),
    City VARCHAR(50)
);
```

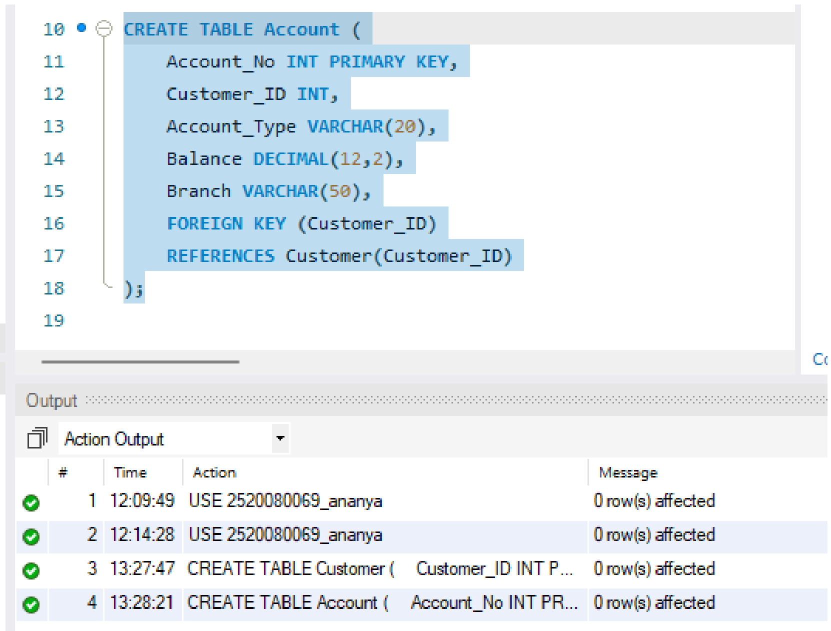


==

3. CREATE ACCOUNT TABLE

==

```
CREATE TABLE Account (
    Account_No INT PRIMARY KEY,
    Customer_ID INT,
    Account_Type VARCHAR(20),
    Balance DECIMAL(12,2),
    Branch VARCHAR(50),
    FOREIGN KEY (Customer_ID)
    REFERENCES Customer(Customer_ID)
);
```

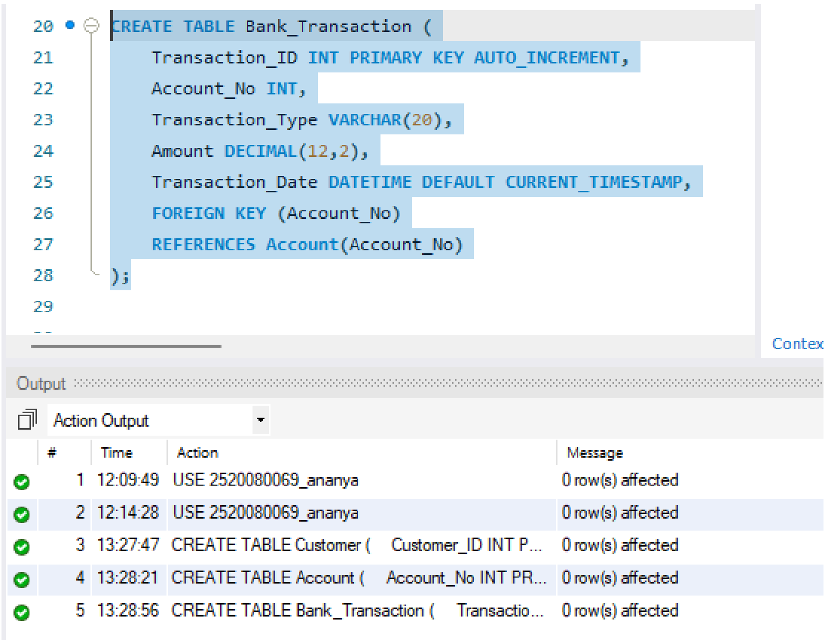


==

4. CREATE TRANSACTION TABLE

==

```
CREATE TABLE Bank_Transaction (
   Transaction_ID INT PRIMARY KEY AUTO_INCREMENT,
   Account_No INT,
   Transaction_Type VARCHAR(20),
   Amount DECIMAL(12,2),
   Transaction_Date DATETIME DEFAULT CURRENT_TIMESTAMP,
   FOREIGN KEY (Account_No)
   REFERENCES Account(Account_No)
);
```

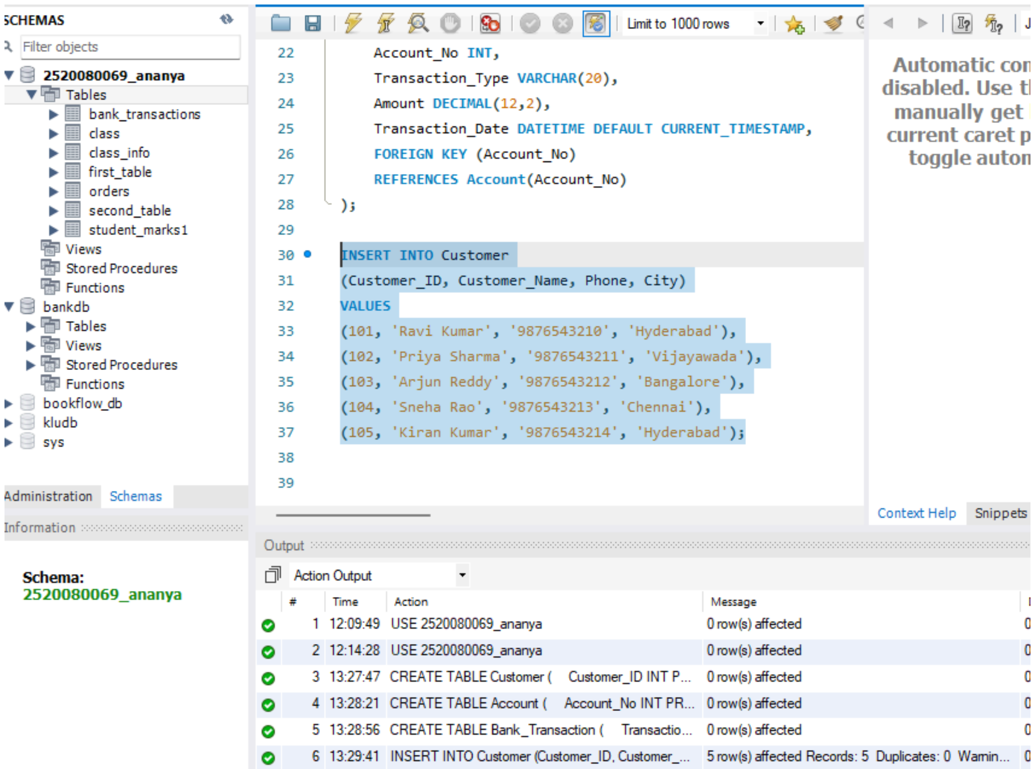


==

5. INSERT CUSTOMER DATA

==

INSERT INTO Customer
(Customer_ID, Customer_Name, Phone, City)
VALUES
(101, 'Ravi Kumar', '9876543210', 'Hyderabad'),
(102, 'Priya Sharma', '9876543211', 'Vijayawada'),
(103, 'Arjun Reddy', '9876543212', 'Bangalore'),
(104, 'Sneha Rao', '9876543213', 'Chennai'),
(105, 'Kiran Kumar', '9876543214', 'Hyderabad');

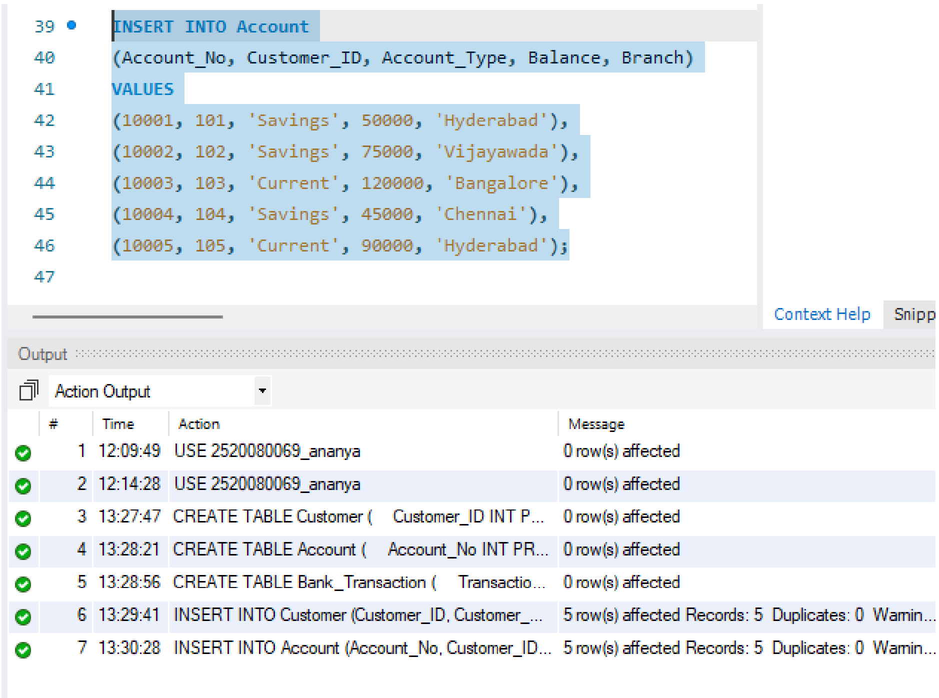


==

6. INSERT ACCOUNT DATA

==

INSERT INTO Account
(Account_No, Customer_ID, Account_Type, Balance, Branch)
VALUES
(10001, 101, 'Savings', 50000, 'Hyderabad'),
(10002, 102, 'Savings', 75000, 'Vijayawada'),
(10003, 103, 'Current', 120000, 'Bangalore'),
(10004, 104, 'Savings', 45000, 'Chennai'),
(10005, 105, 'Current', 90000, 'Hyderabad');

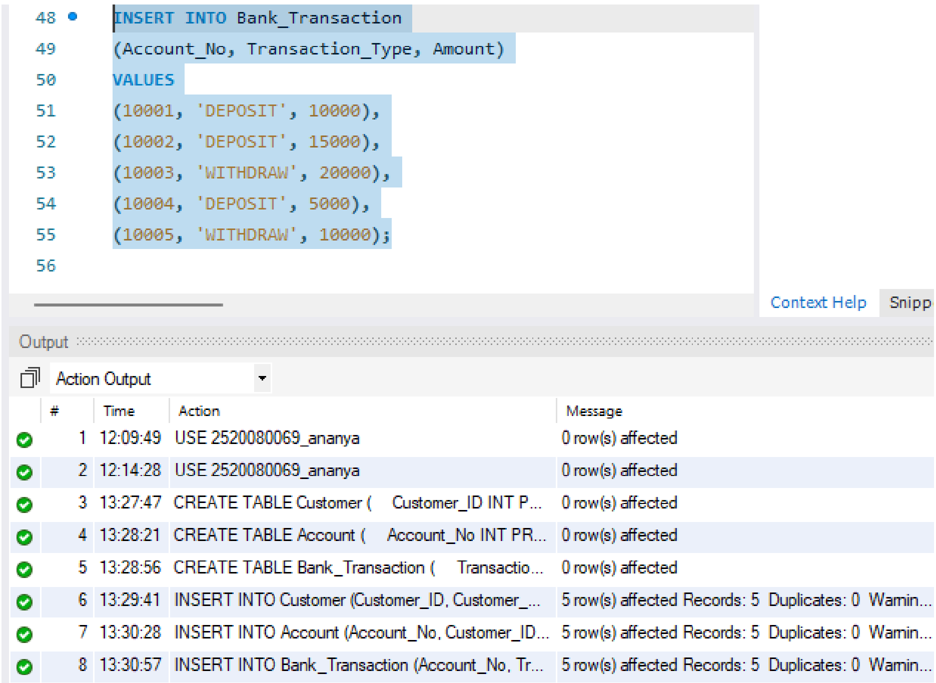


==

7. INSERT BANK TRANSACTION DATA

==

```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);
```



```
INSERT INTO Bank_Transaction
(Account_No, Transaction_Type, Amount)
VALUES
(10001, 'DEPOSIT', 10000),
(10002, 'DEPOSIT', 15000),
(10003, 'WITHDRAW', 20000),
(10004, 'DEPOSIT', 5000),
(10005, 'WITHDRAW', 10000);
```

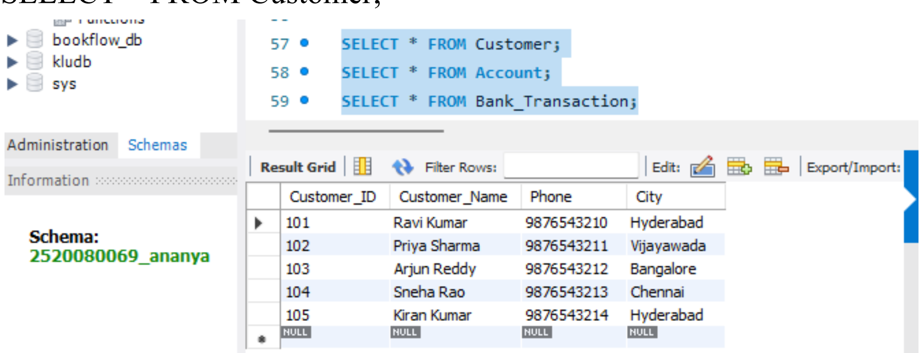

#	Time	Action	Message
1	12:09:49	USE 2520080069_ananya	0 row(s) affected
2	12:14:28	USE 2520080069_ananya	0 row(s) affected
3	13:27:47	CREATE TABLE Customer (Customer_ID INT P...	0 row(s) affected
4	13:28:21	CREATE TABLE Account (Account_No INT PR...	0 row(s) affected
5	13:28:56	CREATE TABLE Bank_Transaction (Transactio...	0 row(s) affected
6	13:29:41	INSERT INTO Customer (Customer_ID, Customer_...	5 row(s) affected Records: 5 Duplicates: 0 Warnin...
7	13:30:28	INSERT INTO Account (Account_No, Customer_ID...	5 row(s) affected Records: 5 Duplicates: 0 Warnin...
8	13:30:57	INSERT INTO Bank_Transaction (Account_No, Tr...	5 row(s) affected Records: 5 Duplicates: 0 Warnin...

==

8. DISPLAY DATA

==

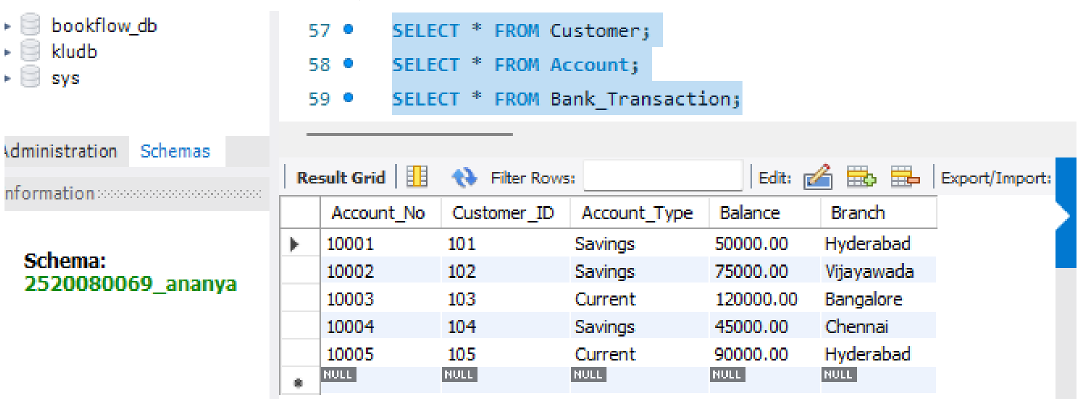
SELECT * FROM Customer;




```
SELECT * FROM Customer;
SELECT * FROM Account;
SELECT * FROM Bank_Transaction;
```


Customer_ID	Customer_Name	Phone	City
101	Ravi Kumar	9876543210	Hyderabad
102	Priya Sharma	9876543211	Vijayawada
103	Arjun Reddy	9876543212	Bangalore
104	Sneha Rao	9876543213	Chennai
105	Kiran Kumar	9876543214	Hyderabad
NULL	NULL	NULL	NULL

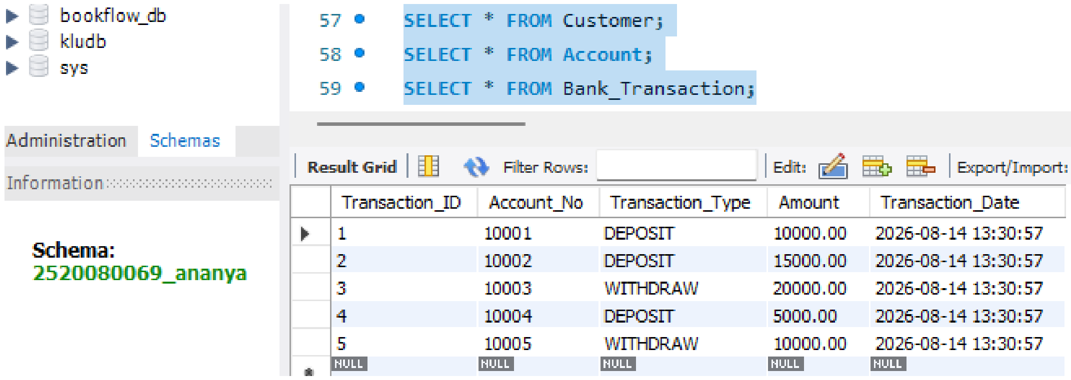
SELECT * FROM Account;




```
SELECT * FROM Customer;
SELECT * FROM Account;
SELECT * FROM Bank_Transaction;
```


Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	50000.00	Hyderabad
10002	102	Savings	75000.00	Vijayawada
10003	103	Current	120000.00	Bangalore
10004	104	Savings	45000.00	Chennai
10005	105	Current	90000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

SELECT * FROM Bank_Transaction;



Transaction_ID	Account_No	Transaction_Type	Amount	Transaction_Date
1	10001	DEPOSIT	10000.00	2026-08-14 13:30:57
2	10002	DEPOSIT	15000.00	2026-08-14 13:30:57
3	10003	WITHDRAW	20000.00	2026-08-14 13:30:57
4	10004	DEPOSIT	5000.00	2026-08-14 13:30:57
5	10005	WITHDRAW	10000.00	2026-08-14 13:30:57
NULL	NULL	NULL	NULL	NULL

==

PART A – TRANSACTIONS

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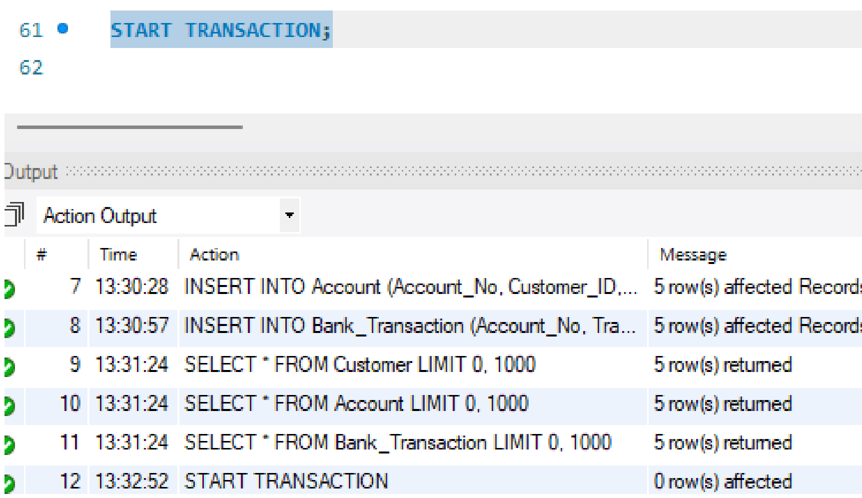
9. START A TRANSACTION

START TRANSACTION;

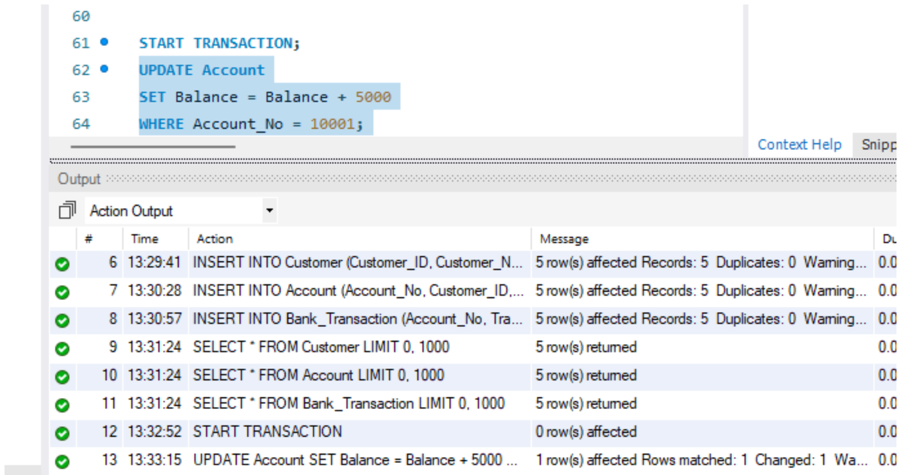


#	Time	Action	Message
7	13:30:28	INSERT INTO Account (Account_No, Customer_ID,...	5 row(s) affected Records
8	13:30:57	INSERT INTO Bank_Transaction (Account_No, Tra...	5 row(s) affected Records
9	13:31:24	SELECT * FROM Customer LIMIT 0, 1000	5 row(s) returned
10	13:31:24	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
11	13:31:24	SELECT * FROM Bank_Transaction LIMIT 0, 1000	5 row(s) returned
12	13:32:52	START TRANSACTION	0 row(s) affected

----

10. DEPOSIT MONEY

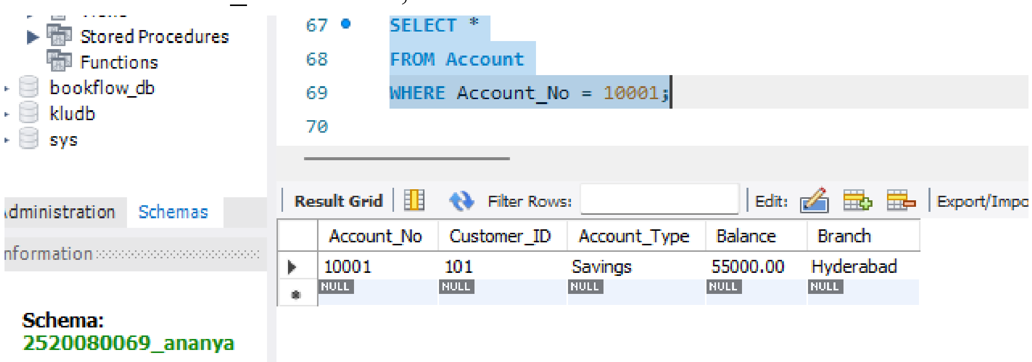
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;



---

11. CHECK BALANCE

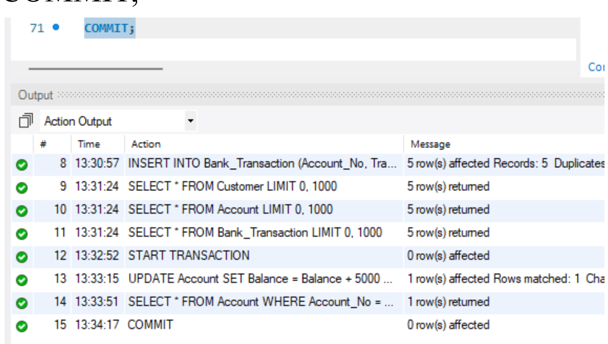
SELECT *
FROM Account
WHERE Account_No = 10001;



---

12. COMMIT TRANSACTION

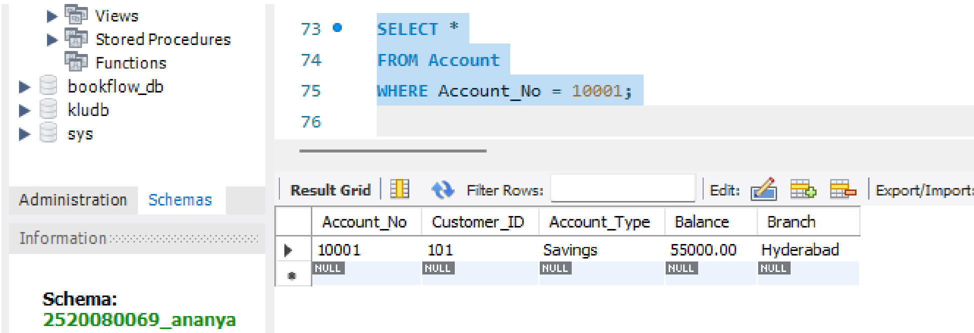
COMMIT;



----

13. CHECK BALANCE AFTER COMMIT

SELECT *
FROM Account
WHERE Account_No = 10001;



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PART B – ROLLBACK

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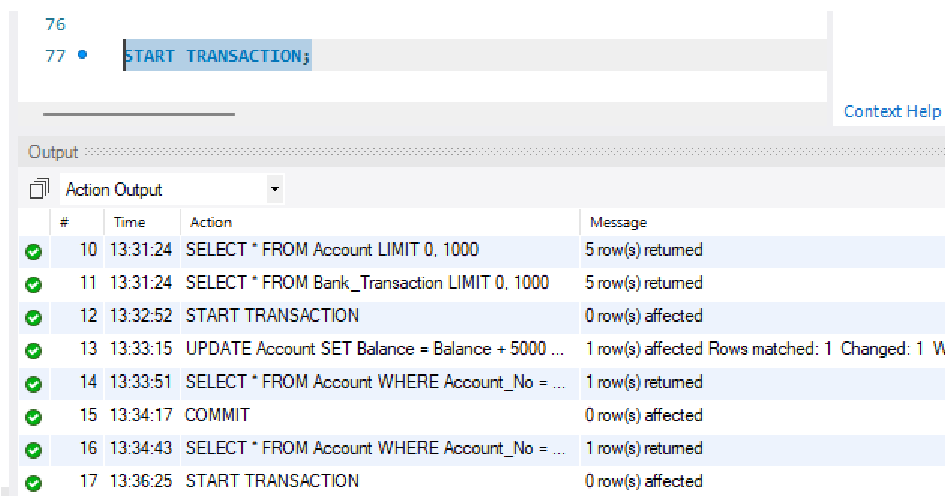
14. START TRANSACTION

START TRANSACTION;

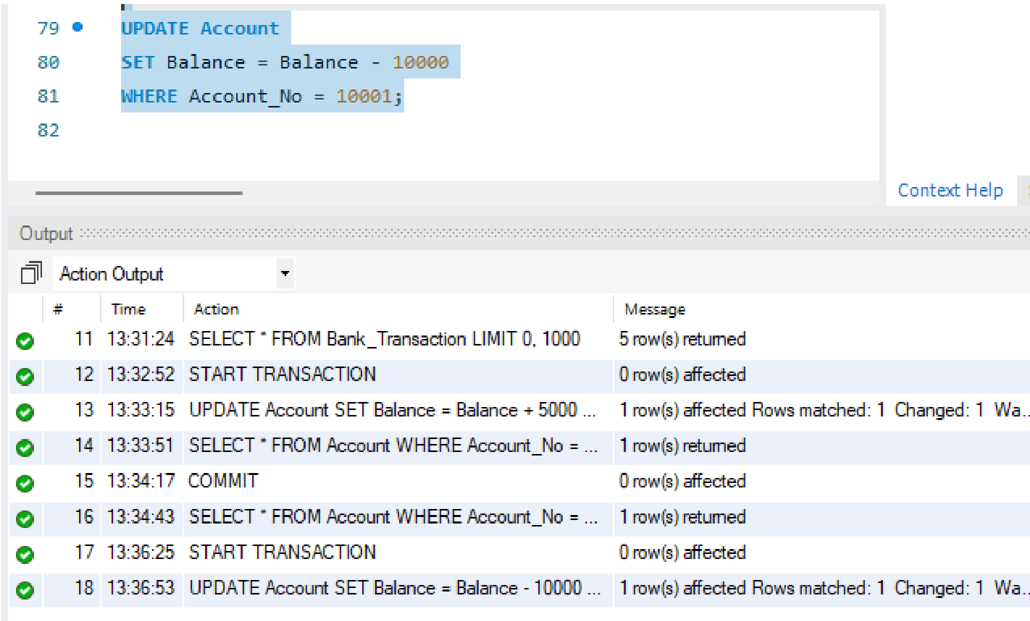


----

15. WITHDRAW MONEY

UPDATE Account
SET Balance = Balance - 10000

WHERE Account_No = 10001;



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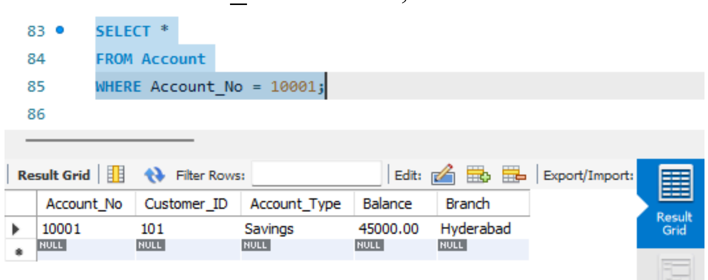
16. CHECK BALANCE

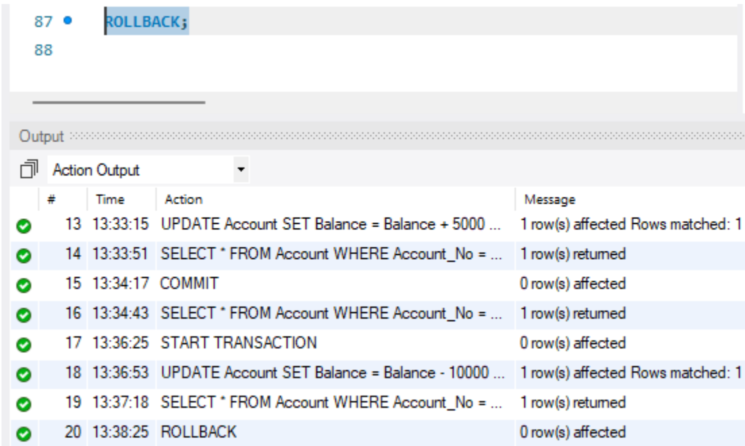
SELECT *
FROM Account
WHERE Account_No = 10001;



----

17. ROLLBACK

ROLLBACK;



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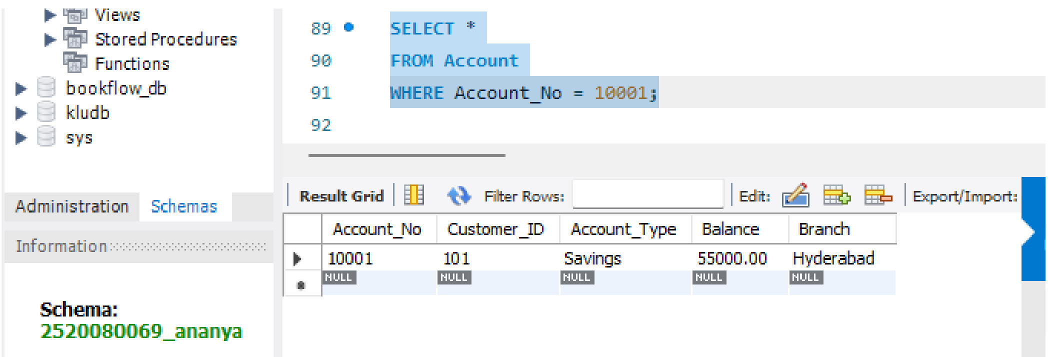
18. CHECK BALANCE AFTER ROLLBACK

```
SELECT *
FROM Account
WHERE Account_No = 10001;
```



===

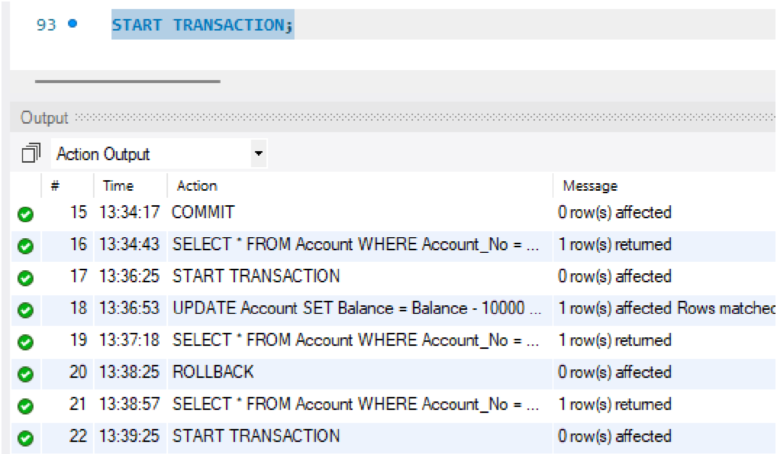
PART C – SAVEPOINT

===

----

19. START TRANSACTION

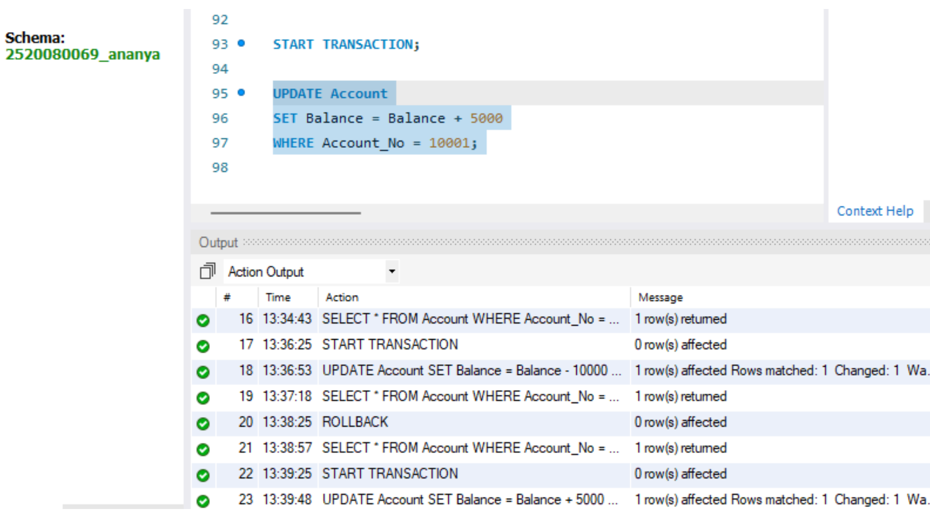
```
START TRANSACTION;
```



---

20. FIRST UPDATE

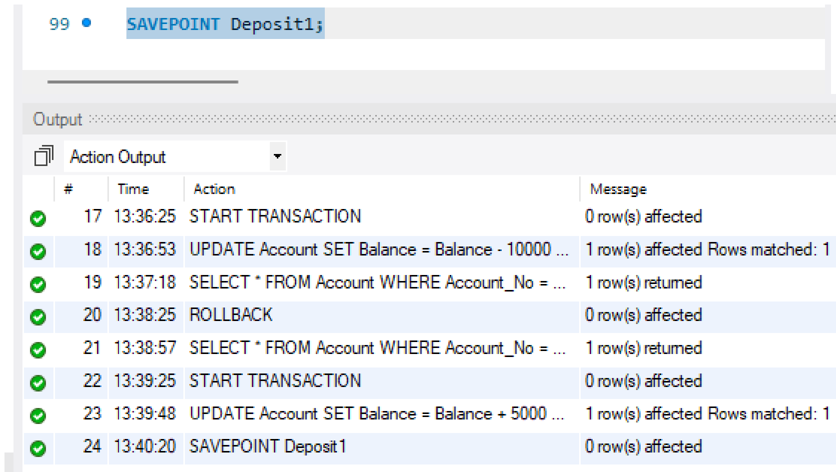
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;



---

21. CREATE SAVEPOINT

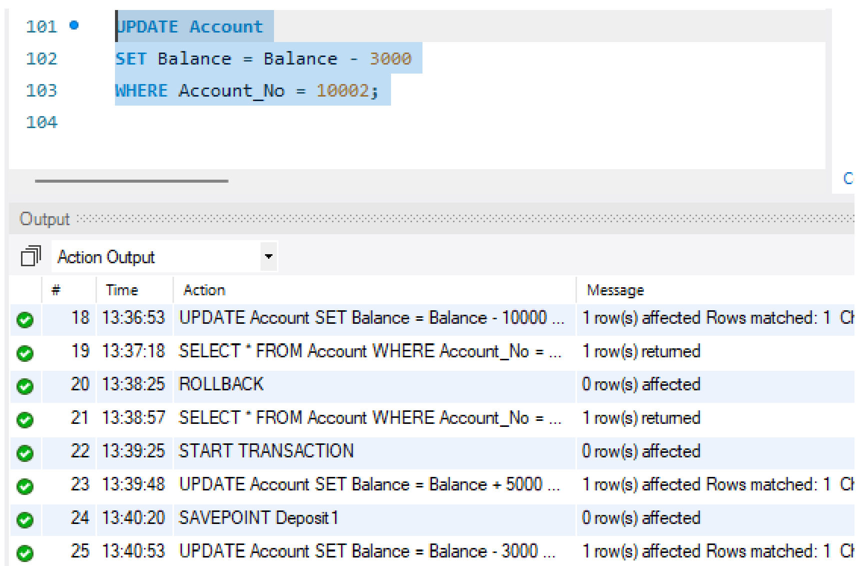
SAVEPOINT Deposit1;



---

22. SECOND UPDATE

UPDATE Account
SET Balance = Balance - 3000
WHERE Account_No = 10002;



---

23. CREATE SECOND SAVEPOINT

SAVEPOINT Withdrawal1;

```sql
SAVEPOINT Withdrawal1;
```

Output

Action Output

#	Time	Action	Message
19	13:37:18	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
20	13:38:25	ROLLBACK	0 row(s) affected
21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
22	13:39:25	START TRANSACTION	0 row(s) affected
23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched:
24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected
25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched:
26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected

---

24. THIRD UPDATE

UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10003;

```sql
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10003;
```

Cor

Output

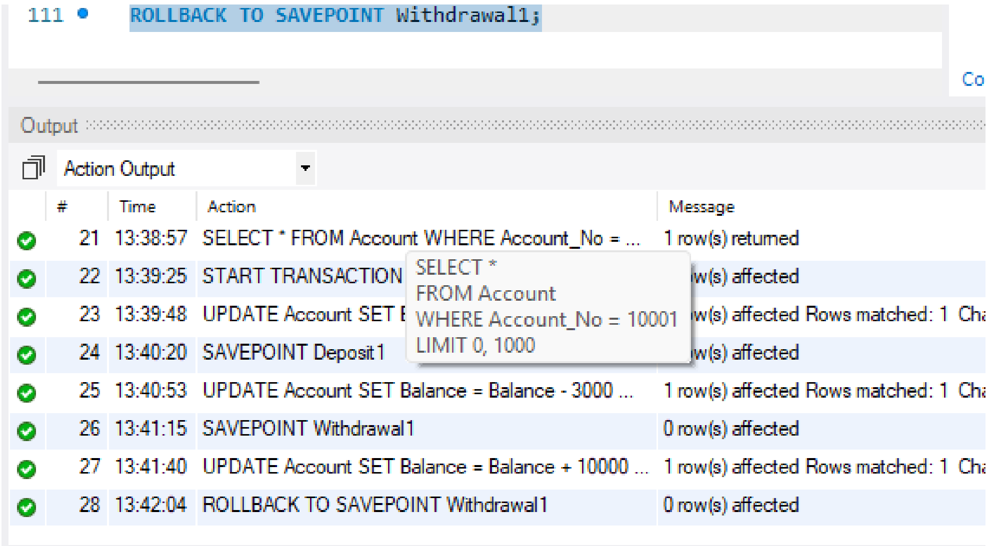
Action Output

#	Time	Action	Message
20	13:38:25	ROLLBACK	0 row(s) affected
21	13:38:57	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
22	13:39:25	START TRANSACTION	0 row(s) affected
23	13:39:48	UPDATE Account SET Balance = Balance + 5000 ...	1 row(s) affected Rows matched: 1 Cha
24	13:40:20	SAVEPOINT Deposit1	0 row(s) affected
25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched: 1 Cha
26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected
27	13:41:40	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Cha

---

25. ROLLBACK TO SAVEPOINT

ROLLBACK TO SAVEPOINT Withdrawal1;



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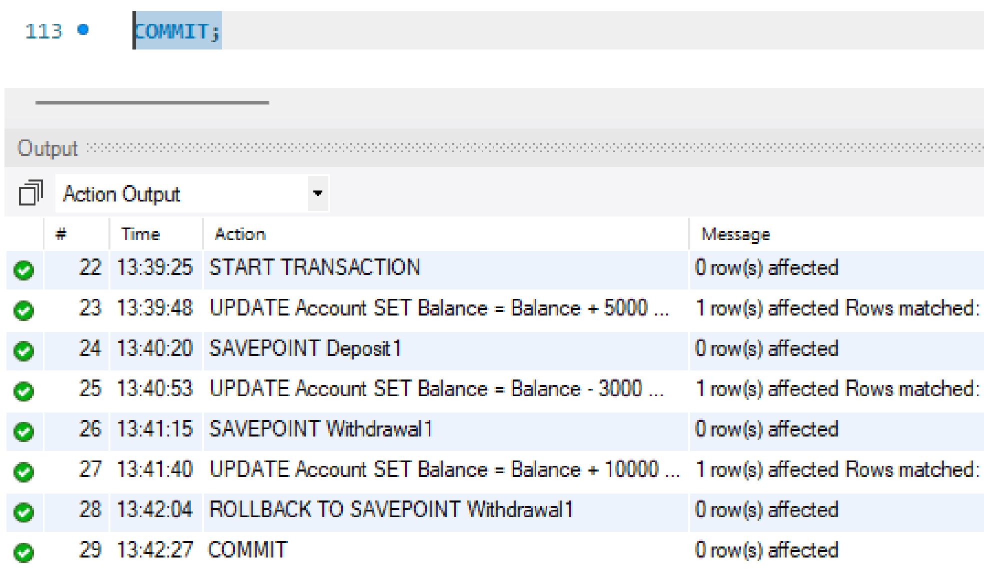
26. COMMIT

COMMIT;



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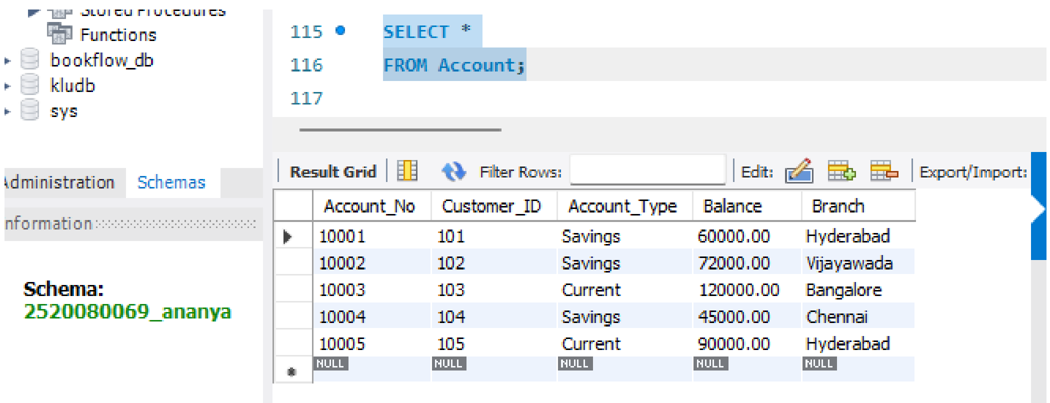
27. CHECK ACCOUNTS

SELECT *
FROM Account;



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PART D – BANK TRANSFER

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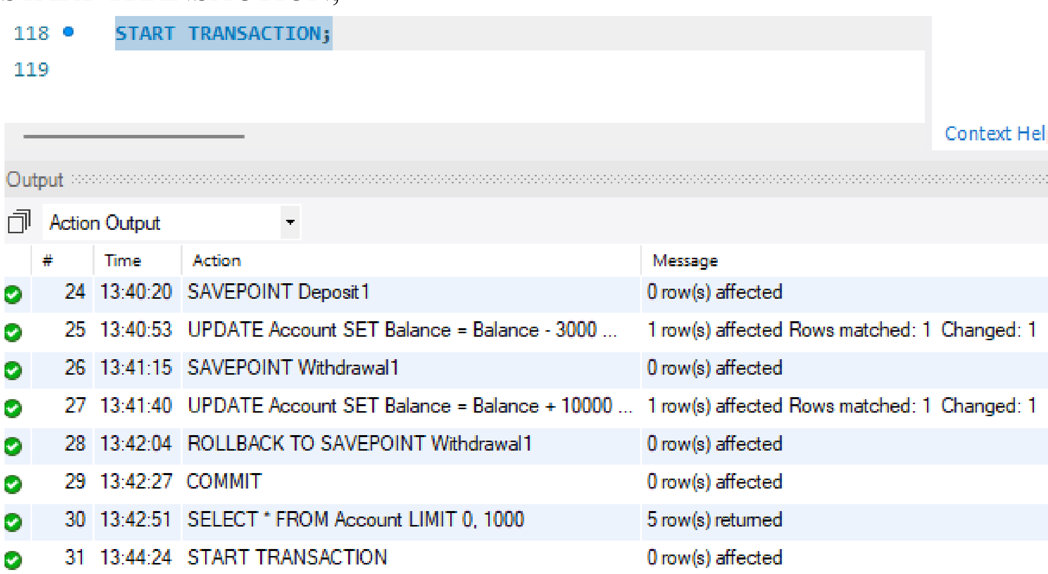
28. TRANSFER MONEY

START TRANSACTION;



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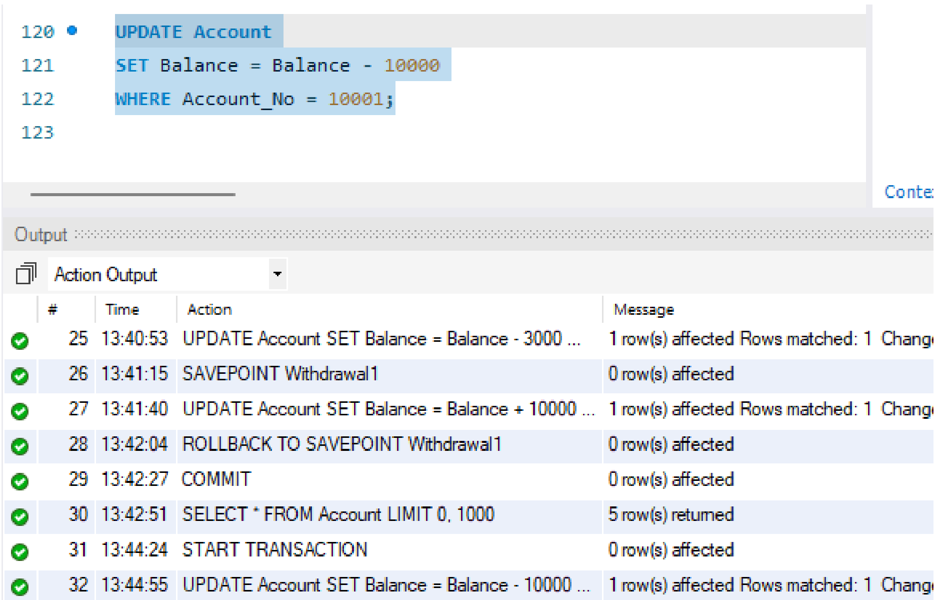
29. DEDUCT FROM SENDER

UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;



```
UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;
```

Conte

Output

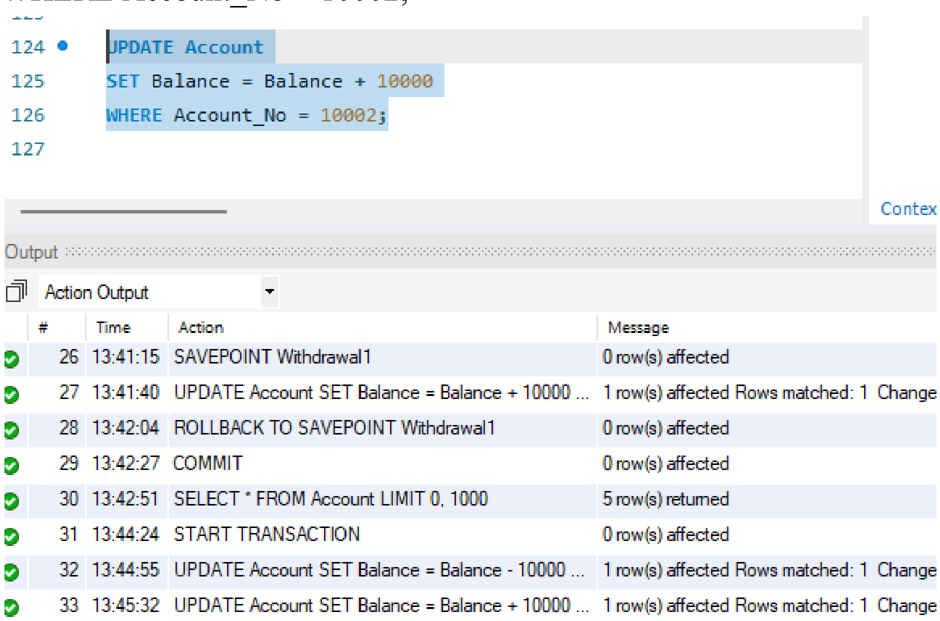
Action Output

#	Time	Action	Message
25	13:40:53	UPDATE Account SET Balance = Balance - 3000 ...	1 row(s) affected Rows matched: 1 Chang
26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected
27	13:41:40	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Chang
28	13:42:04	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
29	13:42:27	COMMIT	0 row(s) affected
30	13:42:51	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
31	13:44:24	START TRANSACTION	0 row(s) affected
32	13:44:55	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Chang

----

30. ADD TO RECEIVER

UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10002;



```
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10002;
```

Contex

Output

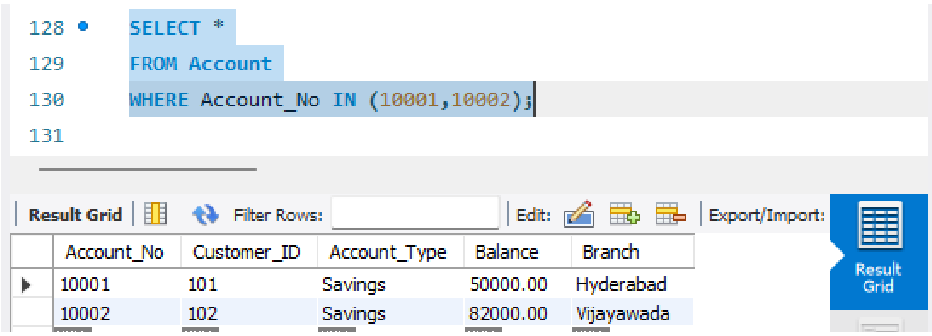
Action Output

#	Time	Action	Message
26	13:41:15	SAVEPOINT Withdrawal1	0 row(s) affected
27	13:41:40	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Change
28	13:42:04	ROLLBACK TO SAVEPOINT Withdrawal1	0 row(s) affected
29	13:42:27	COMMIT	0 row(s) affected
30	13:42:51	SELECT * FROM Account LIMIT 0, 1000	5 row(s) returned
31	13:44:24	START TRANSACTION	0 row(s) affected
32	13:44:55	UPDATE Account SET Balance = Balance - 10000 ...	1 row(s) affected Rows matched: 1 Change
33	13:45:32	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1 Change

----

31. CHECK BOTH ACCOUNTS

```
SELECT *
FROM Account
WHERE Account_No IN (10001,10002);
```



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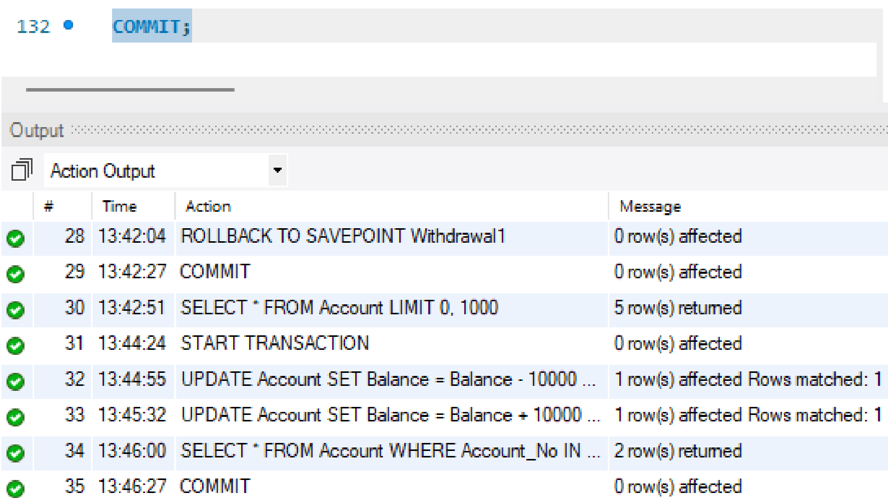
32. COMMIT TRANSFER

```
COMMIT;
```



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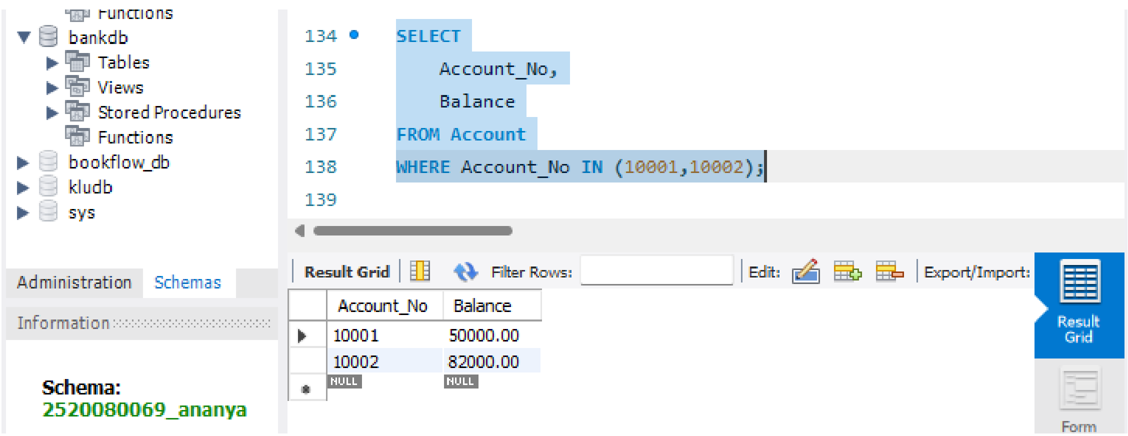
33. FINAL BALANCES

```
SELECT
    Account_No,
    Balance
FROM Account
WHERE Account_No IN (10001,10002);
```



==

PART E – TRANSFER USING ROLLBACK

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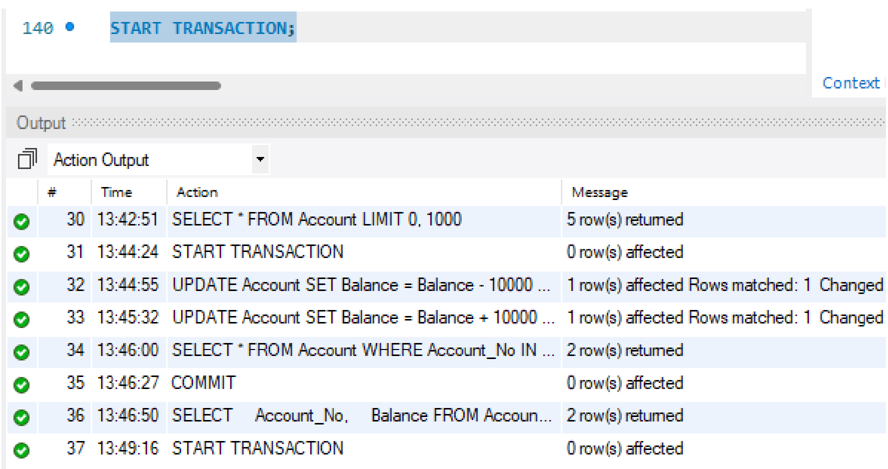
34. START TRANSACTION

START TRANSACTION;



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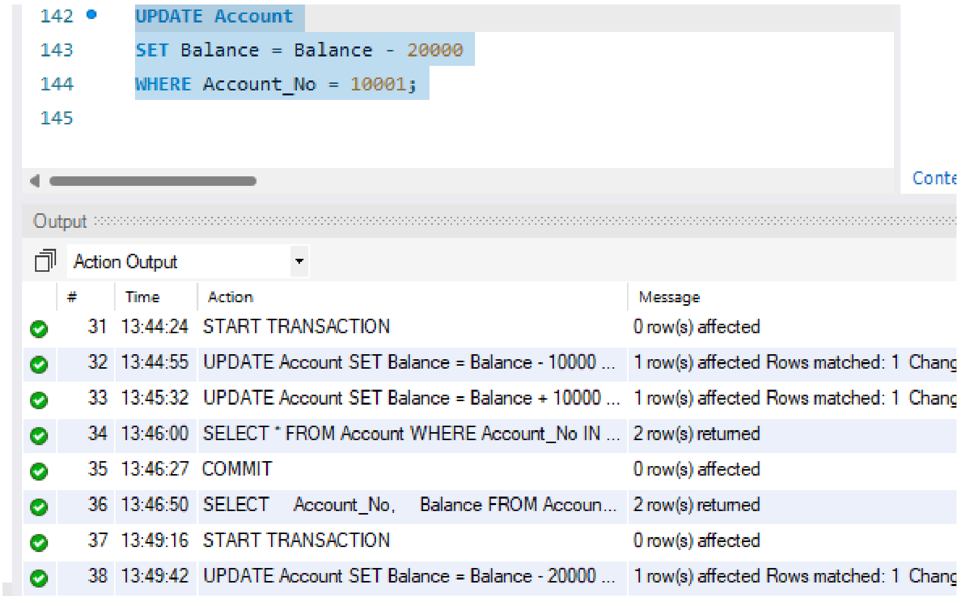
35. DEDUCT MONEY

UPDATE Account
SET Balance = Balance - 20000
WHERE Account_No = 10001;



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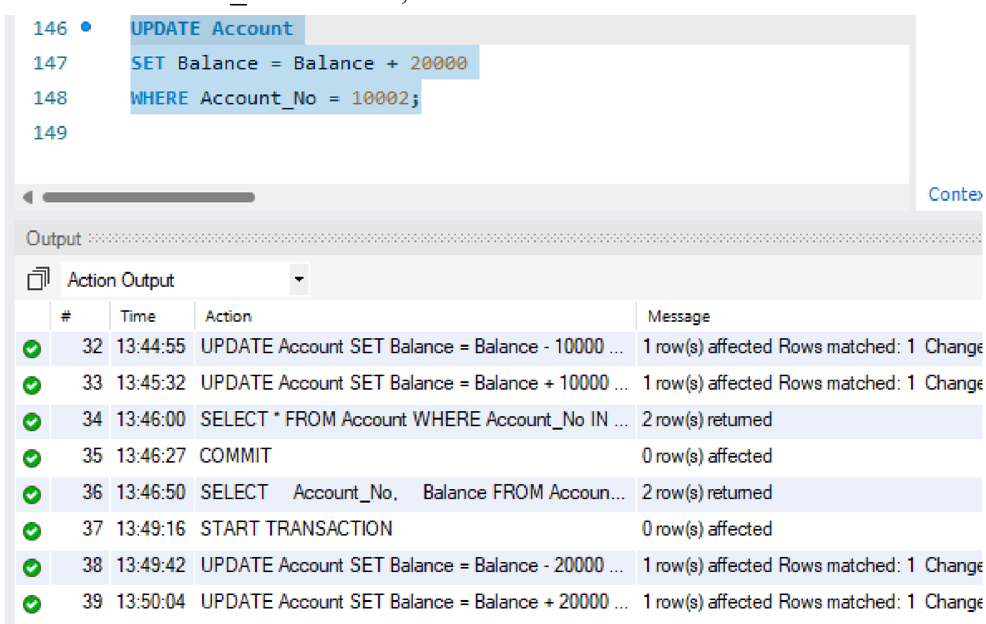
36. ADD MONEY

UPDATE Account
SET Balance = Balance + 20000
WHERE Account_No = 10002;



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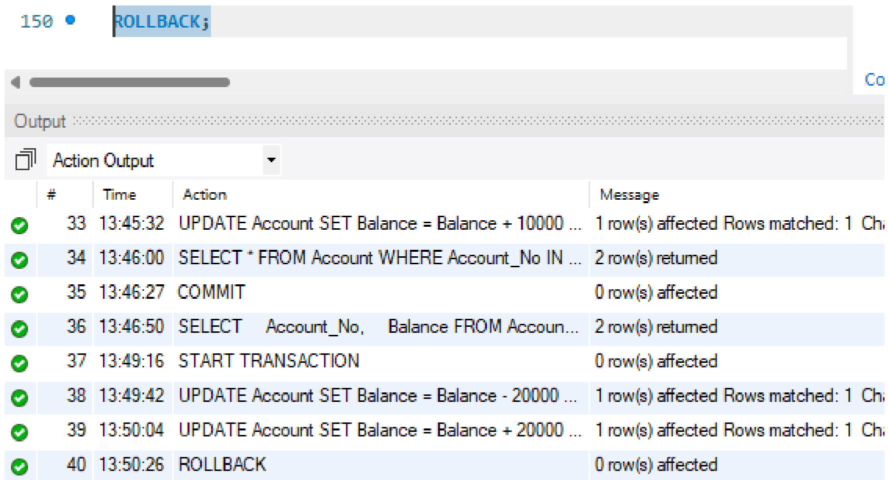
37. CANCEL TRANSACTION

ROLLBACK;



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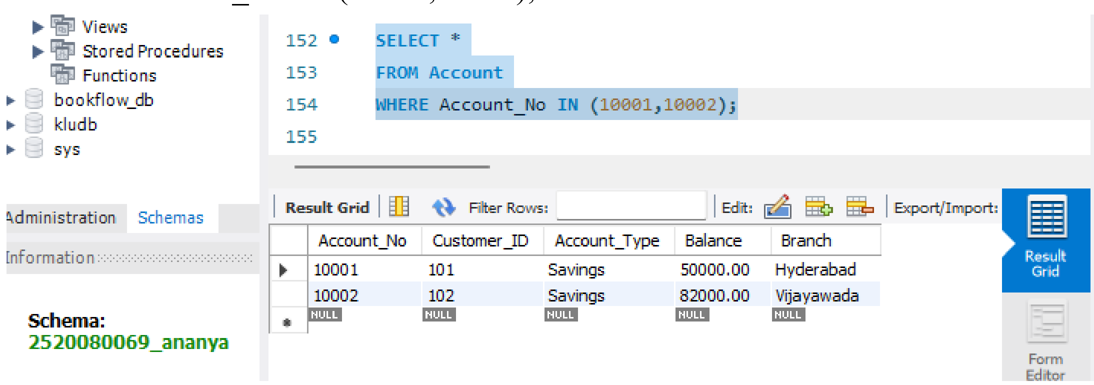
38. CHECK BALANCES

SELECT *
FROM Account
WHERE Account_No IN (10001,10002);



==

PART F – ACID

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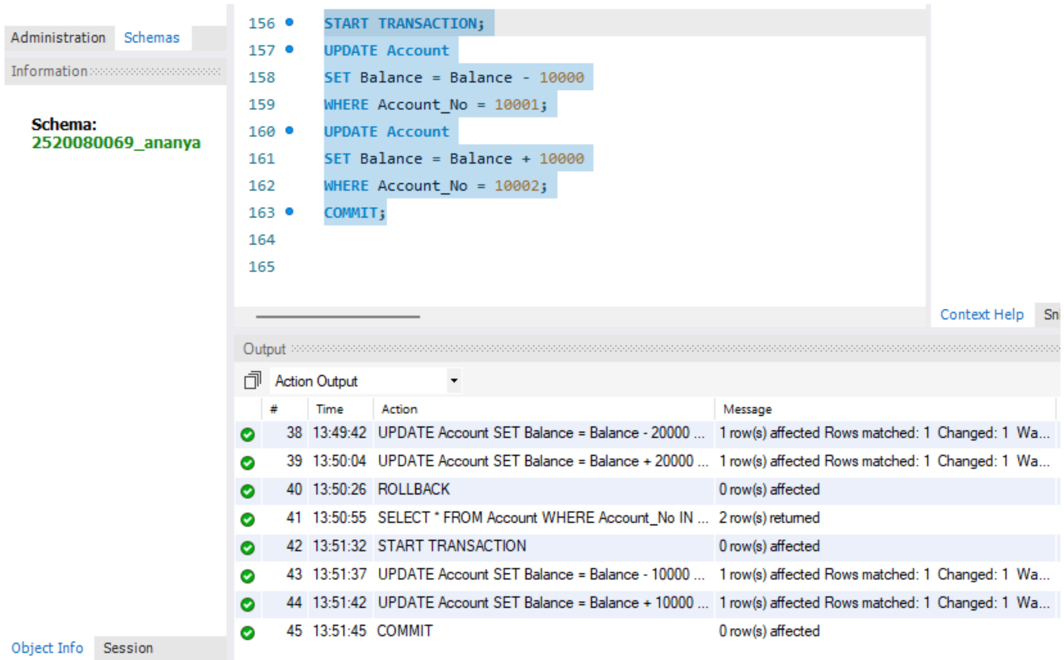
39. ATOMICITY

START TRANSACTION;
UPDATE Account

SET Balance = Balance - 10000
WHERE Account_No = 10001;
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10002;
COMMIT;



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40. CONSISTENCY

Before transfer:
Account 10001 = 50000
Account 10002 = 75000
Total = 125000
After transferring 10000:
Account 10001 = 40000
Account 10002 = 85000
Total = 125000
The total money remains consistent.

----

41. ISOLATION

Isolation means:

Multiple transactions executing at the same time should not incorrectly interfere with each other.

Example:

Transaction T1:

Transfer money.

Transaction T2:

Read account balance.

The database controls how much of T1 is visible to T2.

---

42. DURABILITY

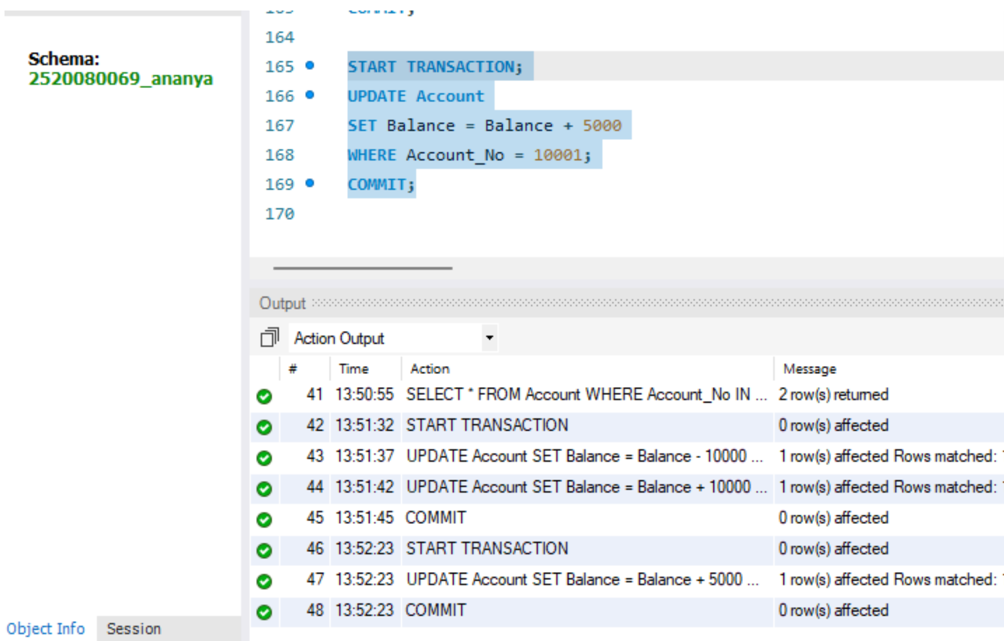
START TRANSACTION;

UPDATE Account

SET Balance = Balance + 5000

WHERE Account_No = 10001;

COMMIT;



==

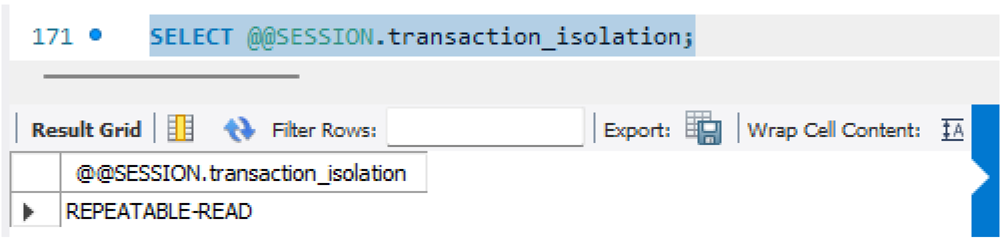
PART G – TRANSACTION ISOLATION LEVELS

==

43. CHECK CURRENT ISOLATION LEVEL

==

SELECT @@SESSION.transaction_isolation;
OR:
SELECT @@GLOBAL.transaction_isolation;



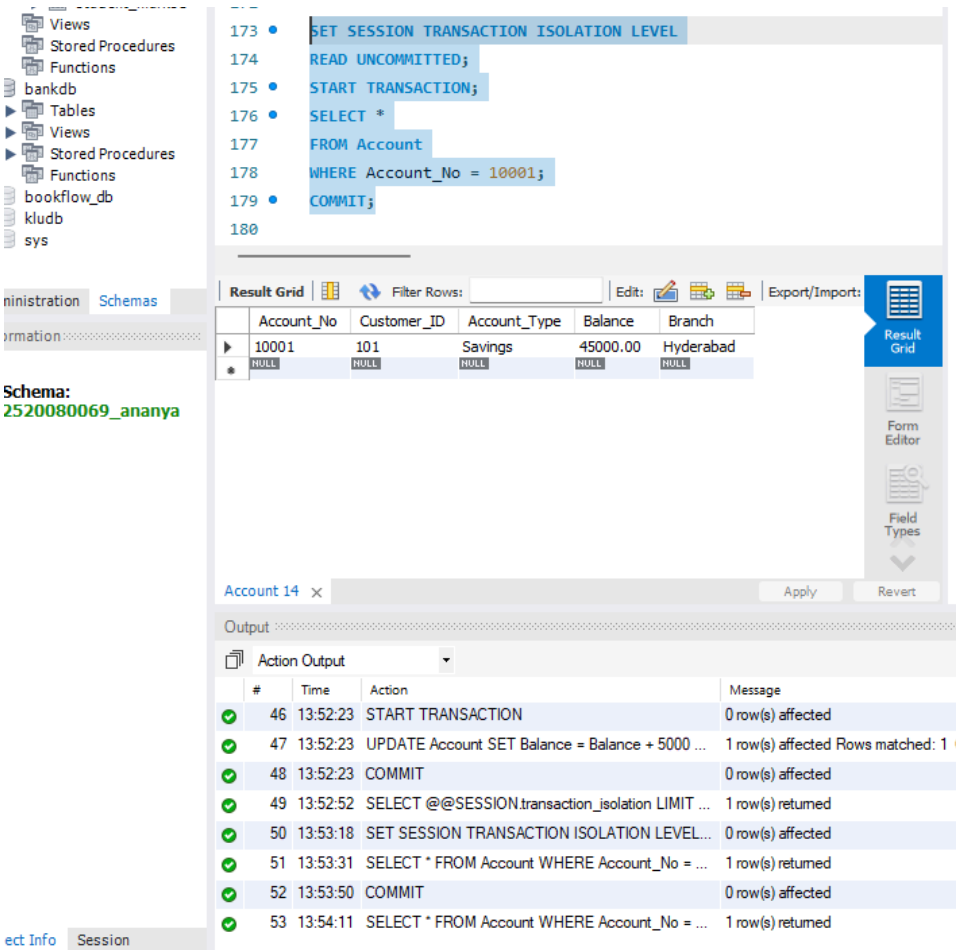
==

44. READ UNCOMMITTED

==

SET SESSION TRANSACTION ISOLATION LEVEL
READ UNCOMMITTED;

START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;

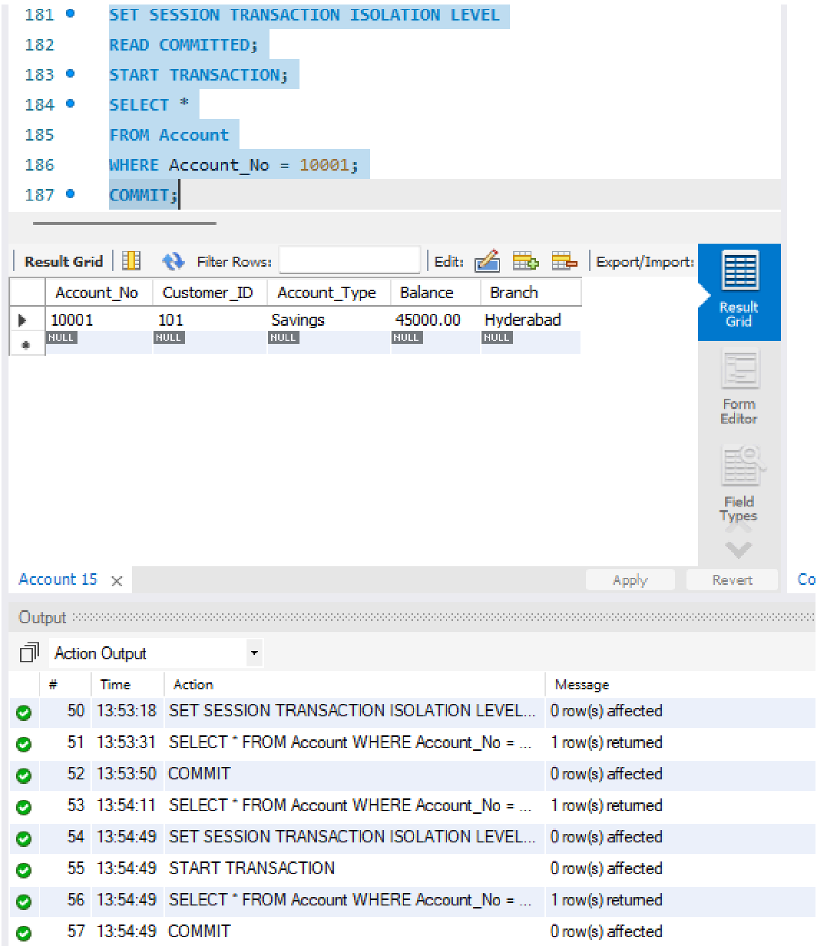


==

45. READ COMMITTED

==

SET SESSION TRANSACTION ISOLATION LEVEL
READ COMMITTED;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;

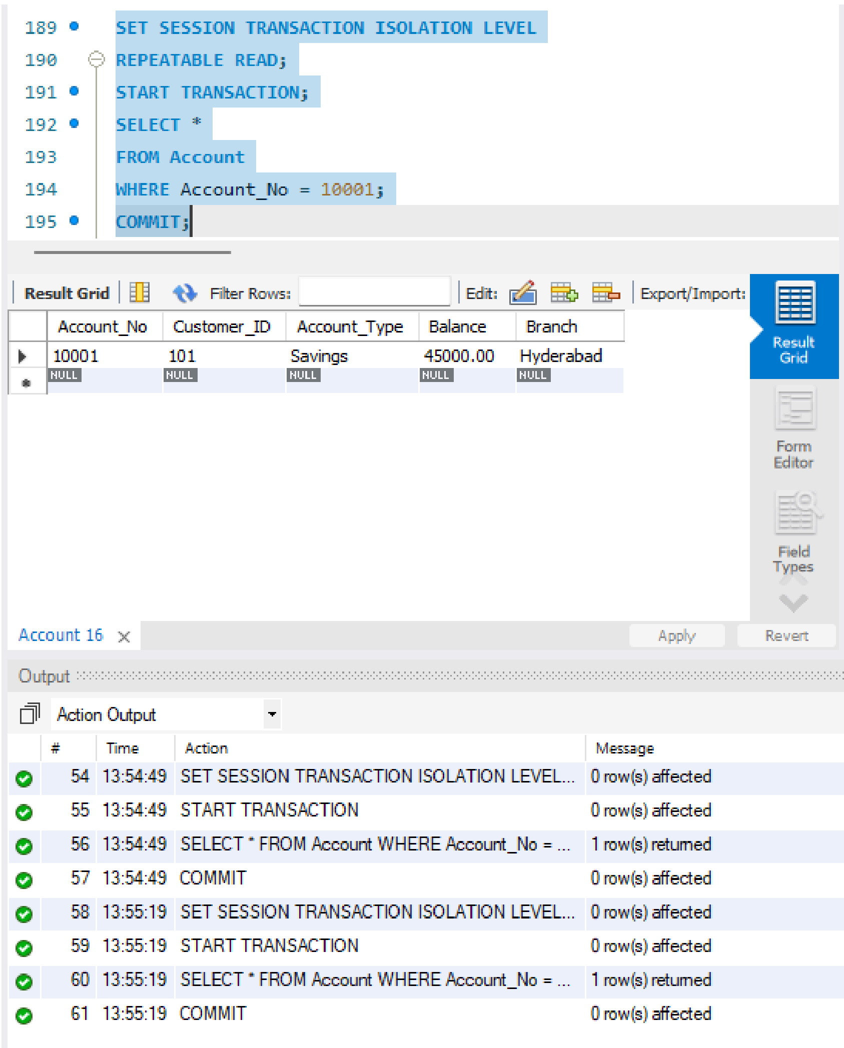


===

46. REPEATABLE READ

===

SET SESSION TRANSACTION ISOLATION LEVEL
REPEATABLE READ;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;



==

47. SERIALIZABLE

==

SET SESSION TRANSACTION ISOLATION LEVEL
SERIALIZABLE;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;

```sql
SET SESSION TRANSACTION ISOLATION LEVEL
SERIALIZABLE;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;
COMMIT;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

Account_No	Customer_ID	Account_Type	Balance	Branch
10001	101	Savings	45000.00	Hyderabad
NULL	NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types

Account 17 × | Apply | Revert

Output

Action Output

#	Time	Action	Message
58	13:55:19	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
59	13:55:19	START TRANSACTION	0 row(s) affected
60	13:55:19	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
61	13:55:19	COMMIT	0 row(s) affected
62	13:55:46	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
63	13:55:46	START TRANSACTION	0 row(s) affected
64	13:55:46	SELECT * FROM Account WHERE Account_No = ...	1 row(s) returned
65	13:55:46	COMMIT	0 row(s) affected

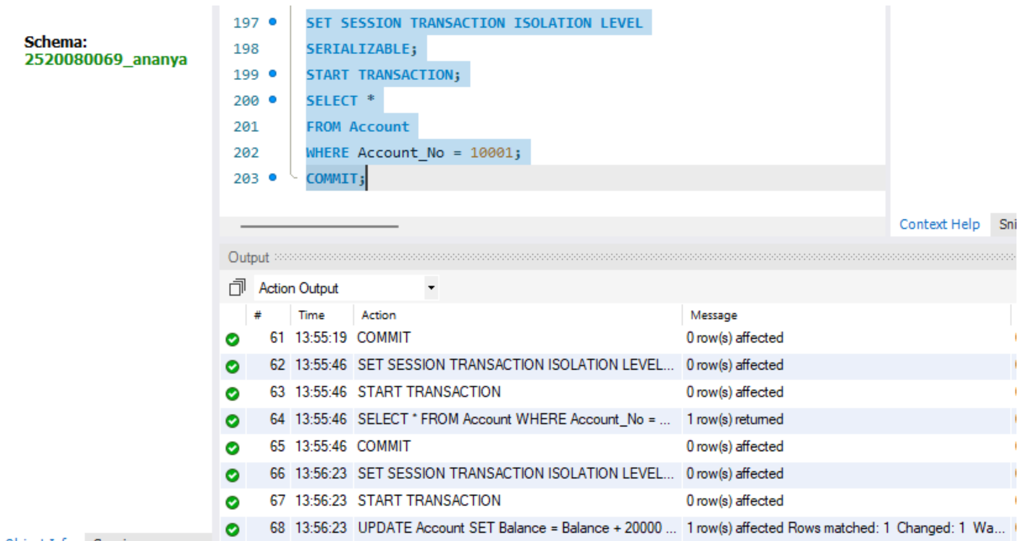
==

PART H – DIRTY READ EXPERIMENT

==

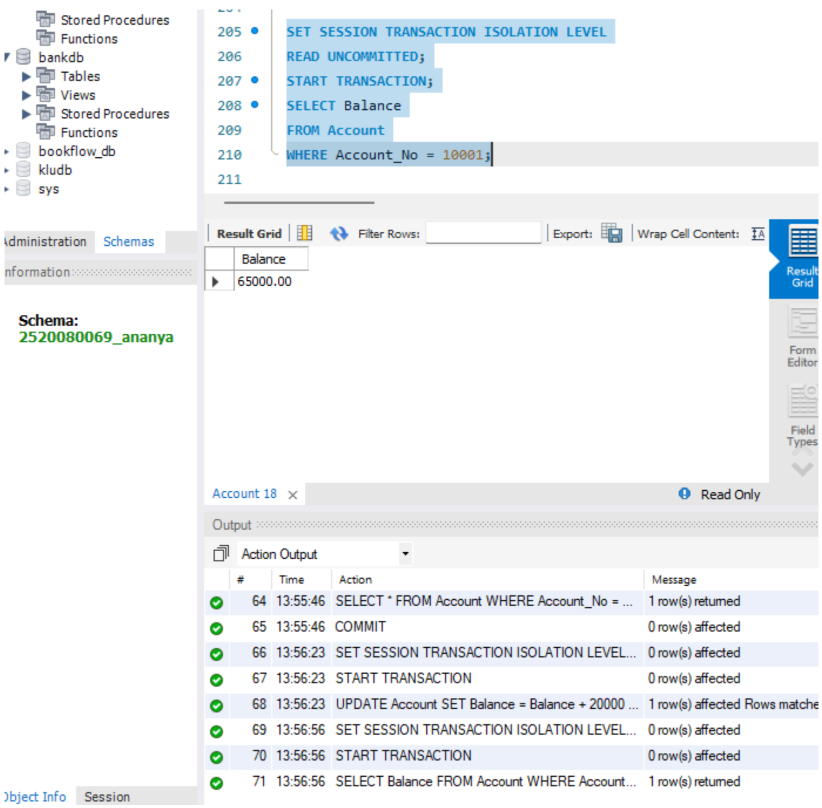
SESSION 1:

SET SESSION TRANSACTION ISOLATION LEVEL
READ UNCOMMITTED;
START TRANSACTION;
UPDATE Account
SET Balance = Balance + 20000
WHERE Account_No = 10001;



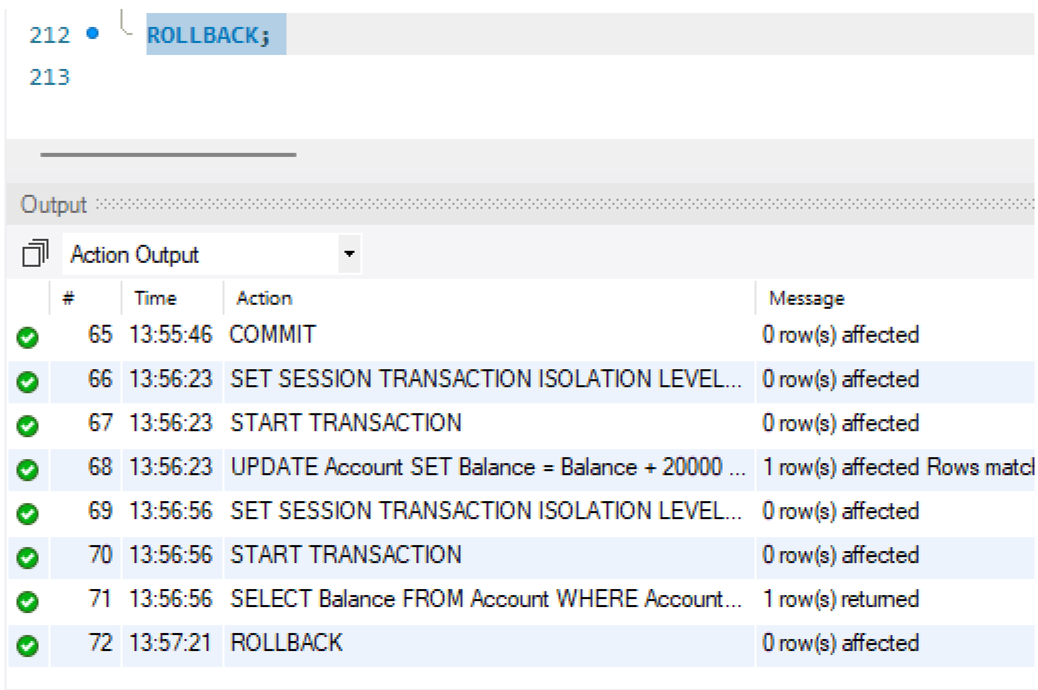
SESSION 2:

SET SESSION TRANSACTION ISOLATION LEVEL
READ UNCOMMITTED;
START TRANSACTION;
SELECT Balance
FROM Account
WHERE Account_No = 10001;



SESSION 1:

ROLLBACK;



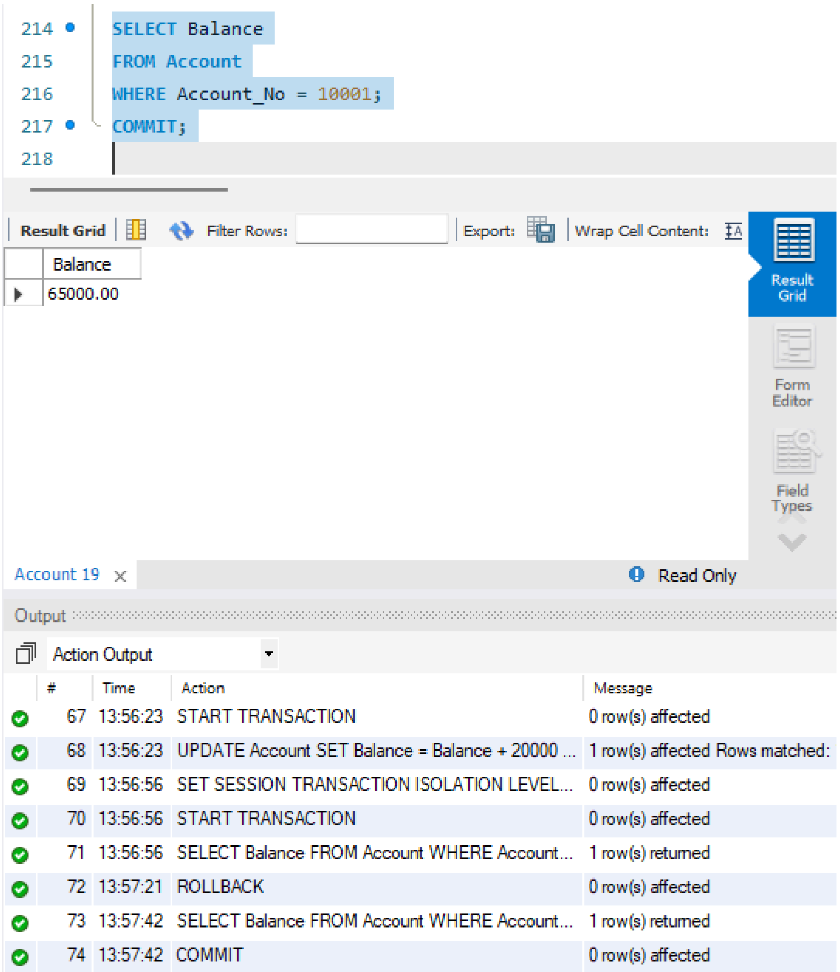
SESSION 2:

SELECT Balance

FROM Account

WHERE Account_No = 10001;

COMMIT;



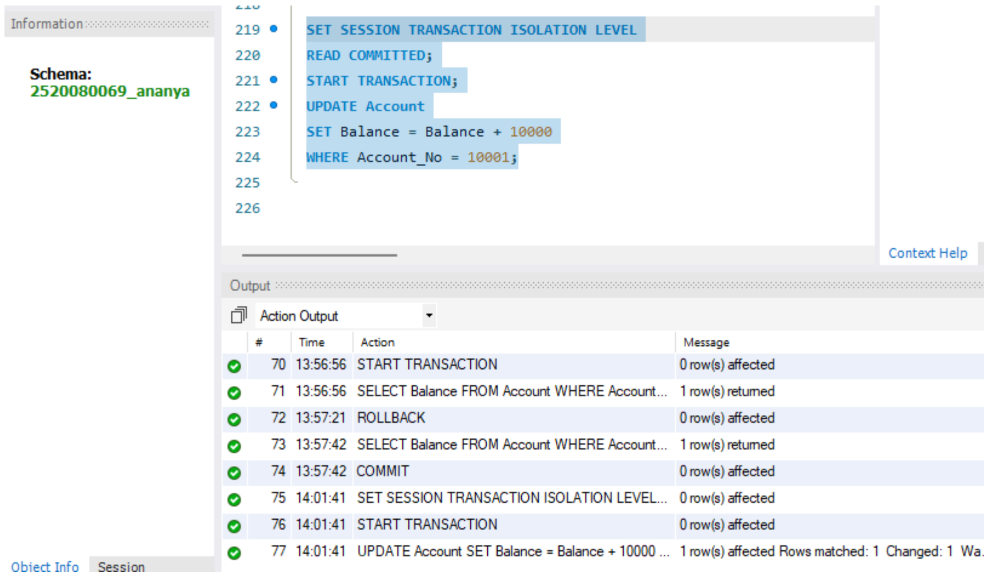
==

PART I – READ COMMITTED EXPERIMENT

==

SESSION 1:

SET SESSION TRANSACTION ISOLATION LEVEL
READ COMMITTED;
START TRANSACTION;
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10001;



SESSION 2:

SET SESSION TRANSACTION ISOLATION LEVEL
READ COMMITTED;
START TRANSACTION;
SELECT Balance
FROM Account
WHERE Account_No = 10001;

```sql
SET SESSION TRANSACTION ISOLATION LEVEL
READ COMMITTED;
START TRANSACTION;
SELECT Balance
FROM Account
WHERE Account_No = 10001;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Balance
75000.00

Account 20 | Read Only

Output

Action Output

#	Time	Action	Message
73	13:57:42	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
74	13:57:42	COMMIT	0 row(s) affected
75	14:01:41	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
76	14:01:41	START TRANSACTION	0 row(s) affected
77	14:01:41	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched:
78	14:02:05	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
79	14:02:05	START TRANSACTION	0 row(s) affected
80	14:02:05	SELECT Balance FROM Account WHERE Account...	1 row(s) returned

SESSION 1:-

COMMIT;

```sql
COMMIT;
```

Output

Action Output

#	Time	Action	Message
74	13:57:42	COMMIT	0 row(s) affected
75	14:01:41	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
76	14:01:41	START TRANSACTION	0 row(s) affected
77	14:01:41	UPDATE Account SET Balance = Balance + 10000 ...	1 row(s) affected Rows matched: 1
78	14:02:05	SET SESSION TRANSACTION ISOLATION LEVEL...	0 row(s) affected
79	14:02:05	START TRANSACTION	0 row(s) affected
80	14:02:05	SELECT Balance FROM Account WHERE Account...	1 row(s) returned
81	14:02:36	COMMIT	0 row(s) affected

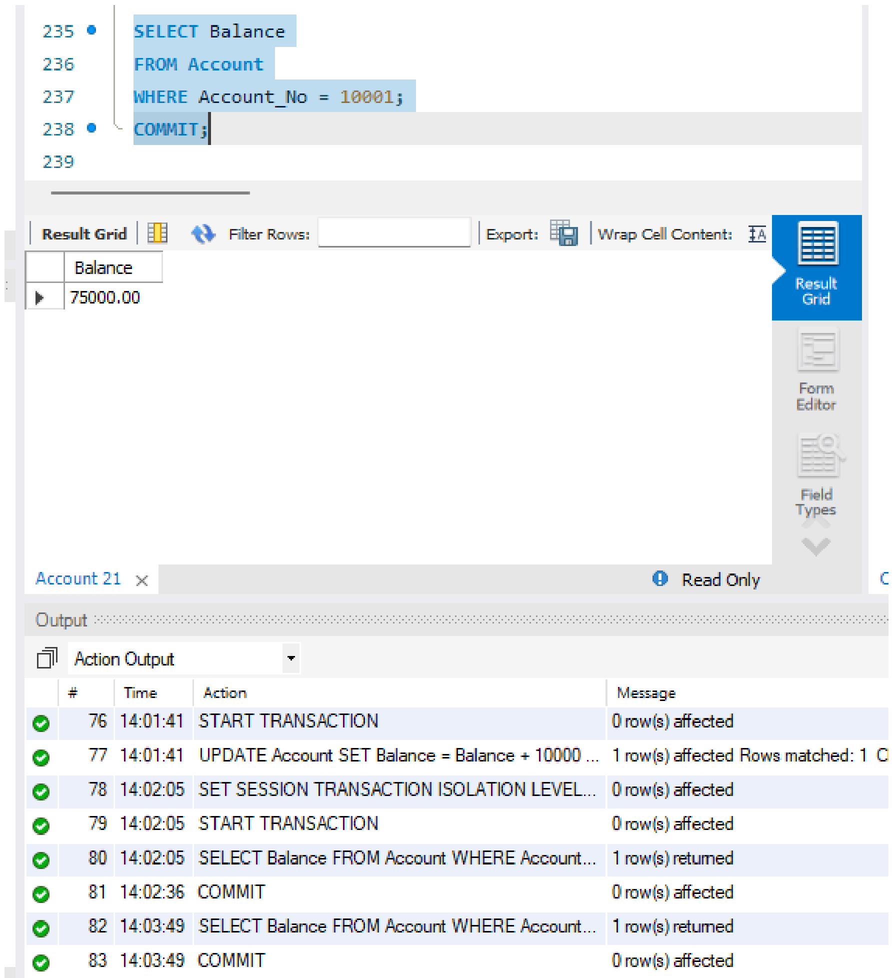
SESSION 2:

SELECT Balance

FROM Account

WHERE Account_No = 10001;

COMMIT;



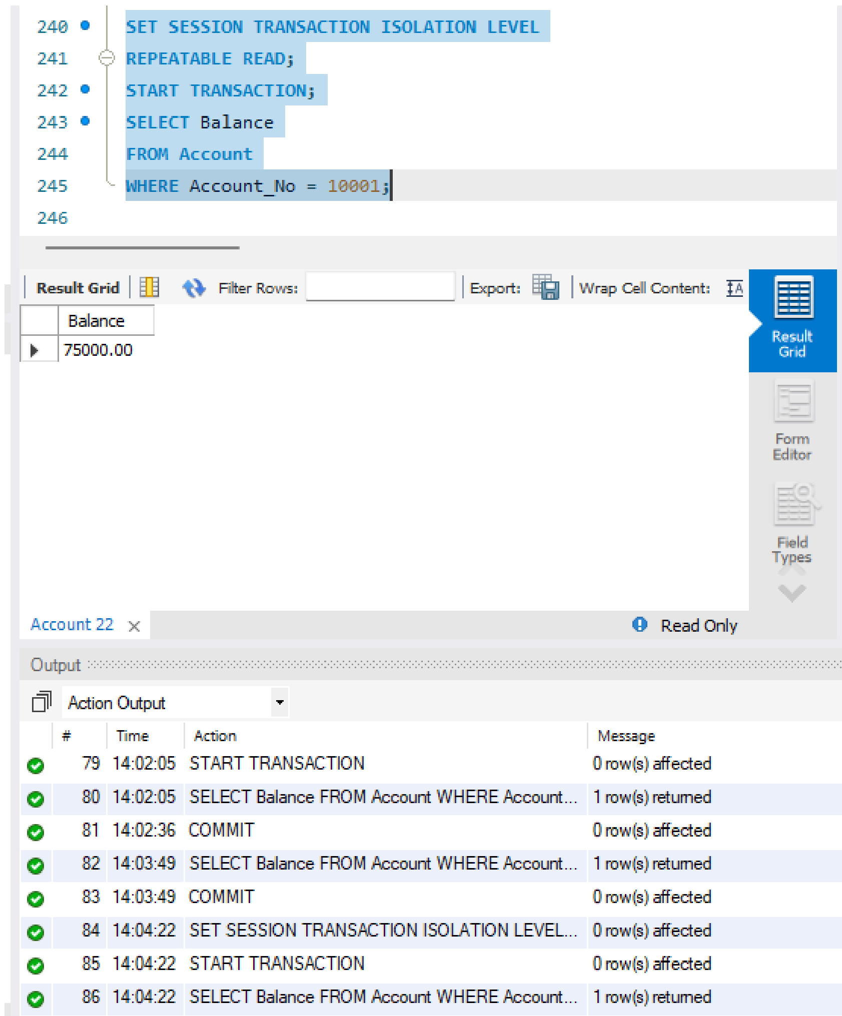
==

PART J – REPEATABLE READ EXPERIMENT

==

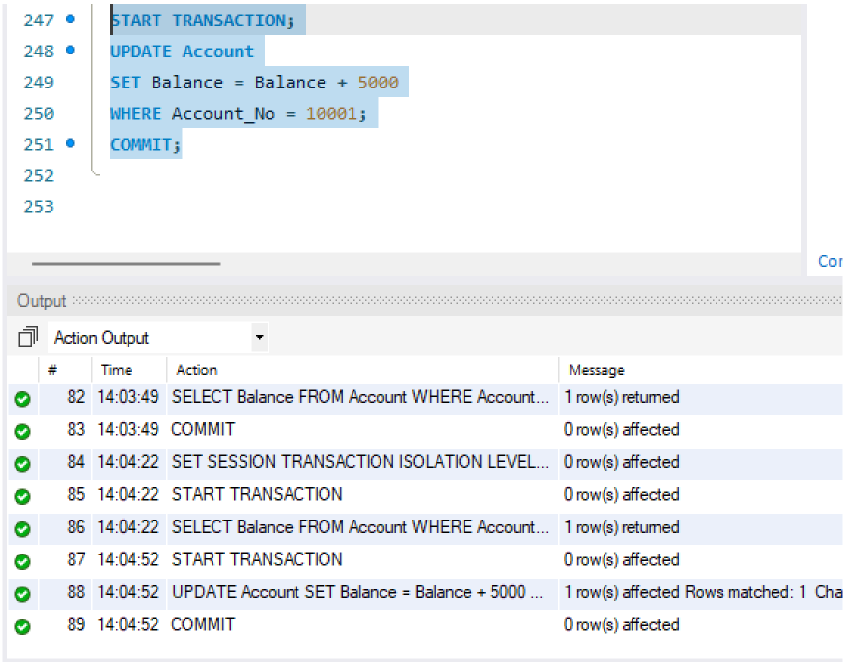
SESSION 1:

SET SESSION TRANSACTION ISOLATION LEVEL
REPEATABLE READ;
START TRANSACTION;
SELECT Balance
FROM Account
WHERE Account_No = 10001;



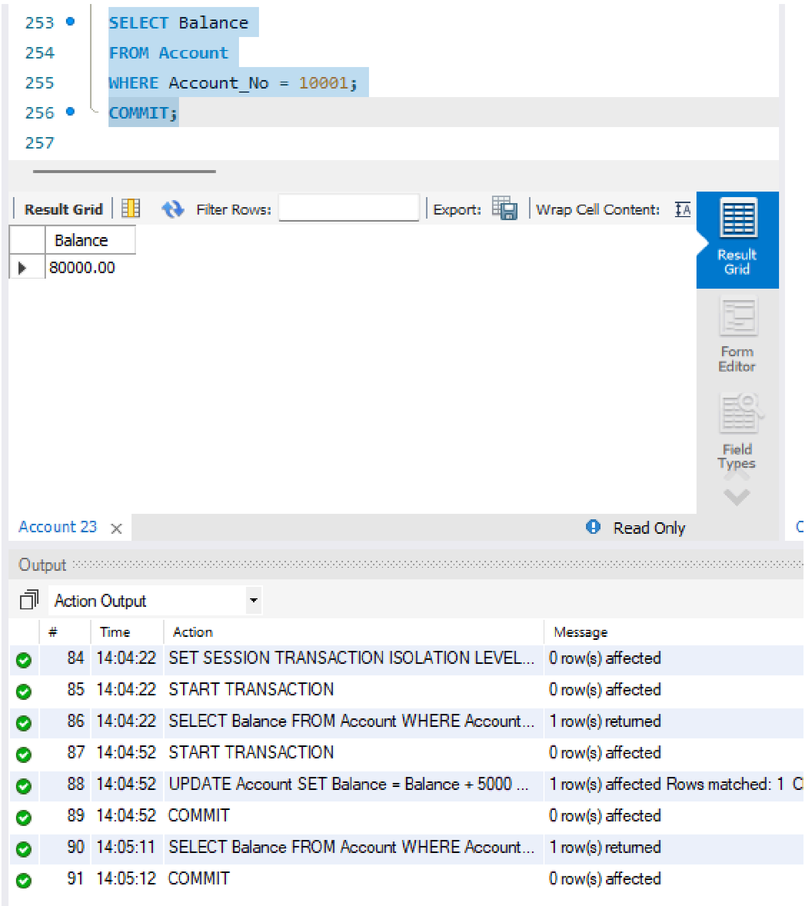
SESSION 2:

START TRANSACTION;
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
COMMIT;



SESSION 1:

SELECT Balance
FROM Account
WHERE Account_No = 10001;
COMMIT;


253 SELECT Balance
254 FROM Account
255 WHERE Account_No = 10001;
256 COMMIT;
257
Result Grid
Filter Rows:
Export:
Wrap Cell Content:
Balance
80000.00
Result Grid
Form Editor
Field Types
Account 23
Read Only
Output
Action Output
Time Action Message
84 14:04:22 SET SESSION TRANSACTION ISOLATION LEVEL... 0 row(s) affected
85 14:04:22 START TRANSACTION 0 row(s) affected
86 14:04:22 SELECT Balance FROM Account WHERE Account... 1 row(s) returned
87 14:04:52 START TRANSACTION 0 row(s) affected
88 14:04:52 UPDATE Account SET Balance = Balance + 5000 ... 1 row(s) affected Rows matched: 1
89 14:04:52 COMMIT 0 row(s) affected
90 14:05:11 SELECT Balance FROM Account WHERE Account... 1 row(s) returned
91 14:05:12 COMMIT 0 row(s) affected

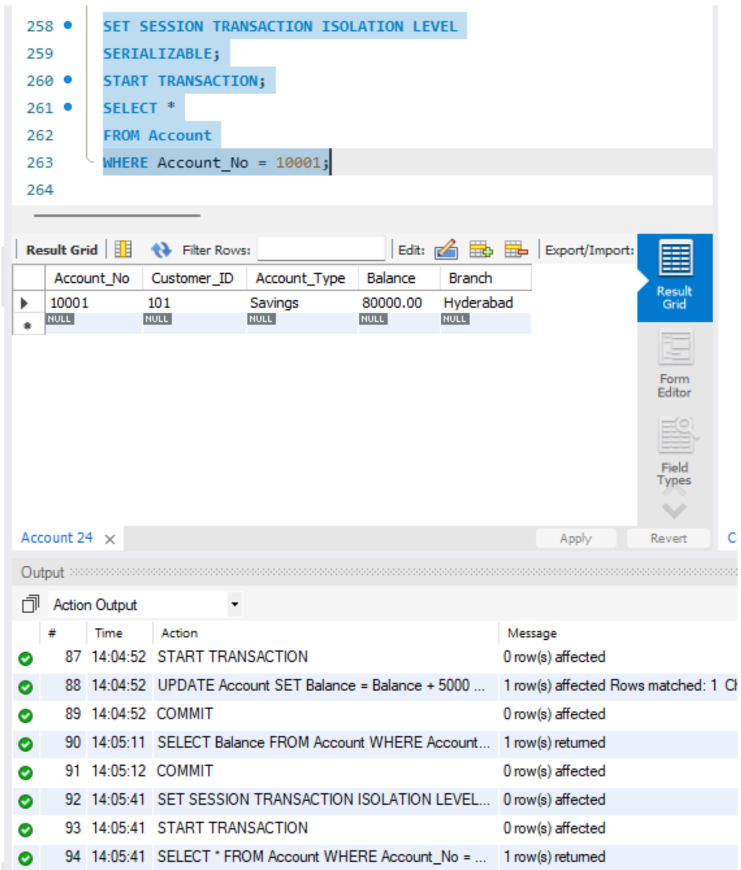
===

PART K – SERIALIZABLE EXPERIMENT

===

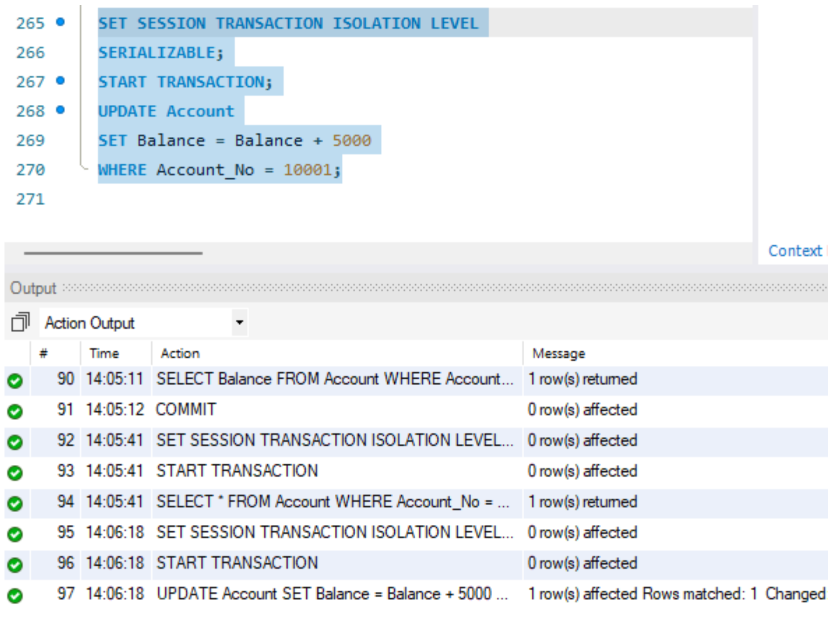
SESSION 1:

SET SESSION TRANSACTION ISOLATION LEVEL
SERIALIZABLE;
START TRANSACTION;
SELECT *
FROM Account
WHERE Account_No = 10001;



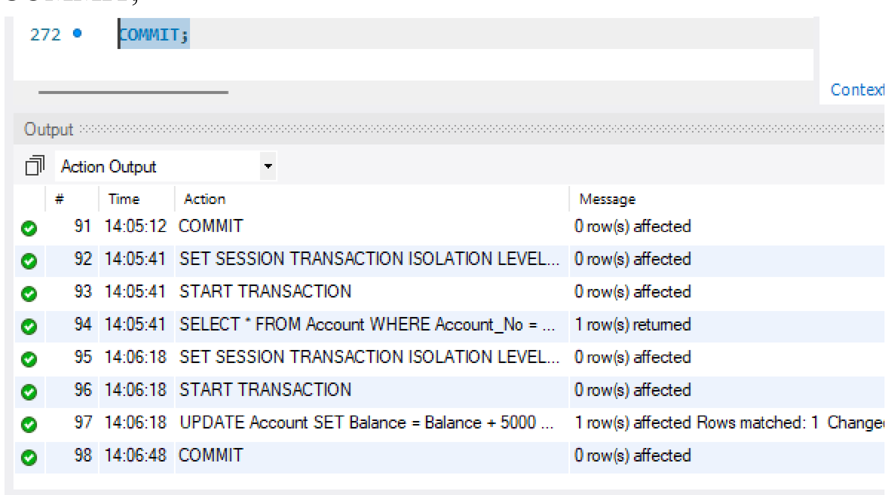
SESSION 2:

SET SESSION TRANSACTION ISOLATION LEVEL
SERIALIZABLE;
START TRANSACTION;
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;



SESSION 1:

COMMIT;



==

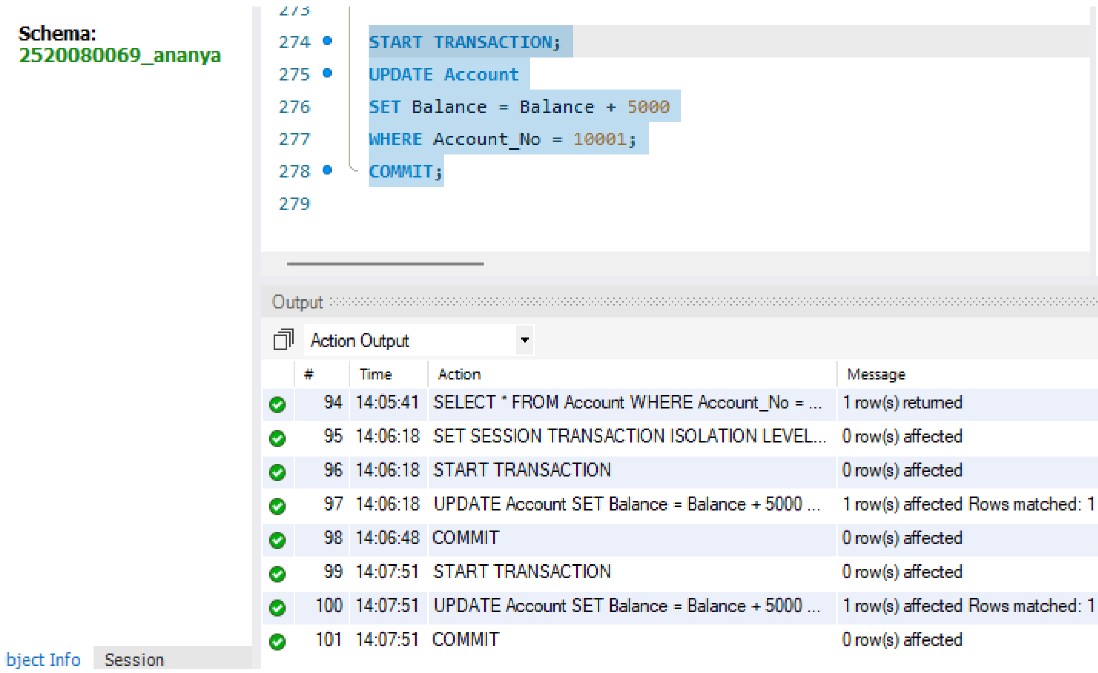
PART M – BANK TRANSACTION EXAMPLES

==

----

48. DEPOSIT TRANSACTION

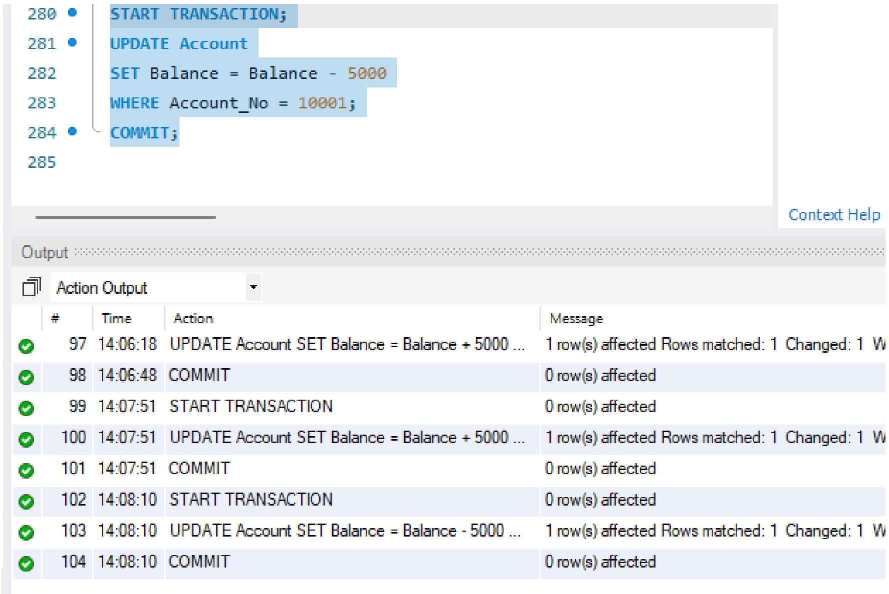
START TRANSACTION;
UPDATE Account
SET Balance = Balance + 5000
WHERE Account_No = 10001;
COMMIT;



----

49. WITHDRAWAL TRANSACTION

START TRANSACTION;
UPDATE Account
SET Balance = Balance - 5000
WHERE Account_No = 10001;
COMMIT;



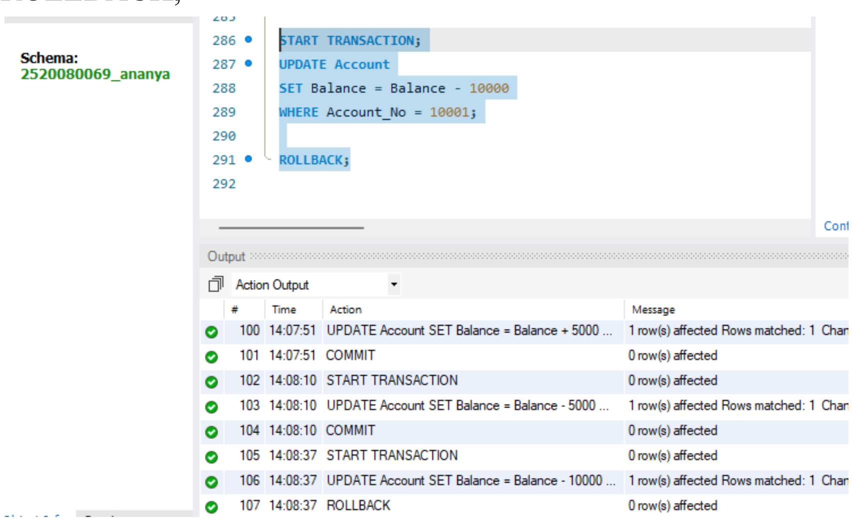
--

50. CANCEL WITHDRAWAL

--

START TRANSACTION;
UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;

ROLLBACK;



----

51. TRANSFER WITH SAVEPOINT

START TRANSACTION;
UPDATE Account
SET Balance = Balance - 10000
WHERE Account_No = 10001;
SAVEPOINT AfterDebit;
UPDATE Account
SET Balance = Balance + 10000
WHERE Account_No = 10002;
COMMIT;